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Microelectrode arrays: from electrophysiology to integrated cancer treatment induced cardiotoxicity assessment

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With ongoing and fast paced progress in cancer chemotherapy and immunotherapy, cancer survival rates have been steadily increased. However, cancer patients, especially in the paediatric and geriatric population are faced with a high risk of cardiotoxicity. This resulted in the emergence of the field of cardio-oncology, and more recently, precision cardio-oncology. A major concern in this context is the development of arrhythmias and contraction dysregulation. These adverse effects can be investigated *in vitro*. In the case of arrhythmia, the Comprehensive *in vitro* Proarrhythmia Assay (CiPA), established in 2013, provides a framework for evaluating proarrhythmic risk with improved specificity and sensitivity compared to traditional assays. CiPA outlines guidelines for preclinical assessment of drug-induced arrhythmia, incorporating a combination of *in vitro*, *in silico*, and clinical approaches applicable to a broad range of pharmaceutical compounds, including chemotherapeutics. The recommended *in vitro* methods under CiPA are transmembrane potential measurement techniques such as patch clamp and automated patch clamp, microelectrode arrays, impedance measurement and the use voltage sensing optical assays (VSOs) such as voltage indicator dyes and genetically encoded voltage indicators (GECIs). CiPA also recommends interfacing these *in vitro* methods with human induced pluripotent stem cells derived cardiomyocytes (hiPSC-CMs) to increase physiological relevance. Currently, there are neither guidelines in place for the *in vitro* assessment of cancer treatment induced arrhythmia (CTIA) or contraction dysregulation, nor any standardized protocols for evaluating cardiotoxicity resulting from cell-based cancer therapies. In this paper we review current *in vitro* approaches for the assessing of cardiotoxic effects of cancer treatments. Putting the focus in the state-of the art use of micro-electrode arrays (MEA) as a novel approach capable of simultaneously monitoring electrophysiological activity, contractile behaviour, and morphological changes induced by chemotherapeutic and immunotherapeutic agents.

KEYWORDS
cardio-oncology, cardiotoxicity, electrophysiology, microelectrode arrays, multimodal

Introduction

Cancer continues to represent a significant global health challenge. In 2020, approximately 20 million new cancer cases were documented, resulting in approximately 10 million cancer-related deaths, representing nearly one in six global fatalities according to GLOBOCAN (Sung et al., 2021). Historically, treatment options for cancer patients were predominantly limited to surgery, radiotherapy and chemotherapy, either as standalone treatments or in combination therapies. Recent improved advancements in understanding of cancer pathophysiology and progression, have revolutionized cancer treatment. New approaches such as drugs, biological molecules, and immune-mediated therapies have emerged. Chemotherapy and immunotherapy are now widely adopted and form the central focus of this review (Liu et al., 2024).

Due to this revolution in cancer treatment, there has been a rise in survival rate of cancer and consequently highlighting the long-term side effects of these treatments, notably cardiotoxicity (Figure 1). The cardiotoxic side effects caused by cancer treatments can manifest as arrhythmia such as QT prolongation, Torsade de Pointes (TdP) or atrial fibrillation which can lead to exacerbated symptoms such as hemodynamic deterioration or even sudden death (Hiraoka Gussak et al., 2008). Other cardiotoxicity side effects are contraction dysregulation which may lead to heart failure (Riccardi et al., 2023; Kistamás et al., 2020; Keefe et al., 2023; Njegic et al., 2020) and structural alterations to heart (Panpan et al., 2022). Early detection, prevention, and management of cancer treatment induced arrhythmia and contractile dysfunctions are crucial to enhancing patient outcomes and quality of life (Lyon et al., 2022; Fradley et al., 2021). The mechanism of cancer treatment induced cardiotoxicity and its manifestation as arrhythmia and contraction dysregulation is still to be firmly established. In most cases, chemotherapeutic induced arrhythmia is manifested as increased QT prolongation (Buza et al., 2017) which has been studied mostly with Electrocardiograms or ECG.

Cardiotoxicity also poses a significant challenge in drug development due to high costs (approximately $2.8 billion) and extended timelines (about 15 years) involved in bringing new drugs to market (Simoens and Huys, 2021). Concerns over cardiotoxicity-related drug attrition underscore the critical need for robust and standardized guidelines for screening cancer-induced cardiotoxicity (CIC) during both the development of new therapies and retrospective evaluations of existing treatments.

The first report of cancer treatment induced cardiotoxicity was in 1966 due to a widely employed class of chemotherapeutic agents (anthracyclines) (Fabiani et al., 2025). This was accompanied by the emergence of a new field, Cardio-oncology (dedicated to understanding and managing cardiovascular complications of cancer treatments). Historically, awareness of drug-induced arrhythmogenic risk increased significantly in the late 20th century; between 1988 and 2009. During this time, 14 drugs were withdrawn from the market worldwide as a result of arrhythmogenic effects, including TdP (Qu et al., 2023). This concern prompted the establishment of the International Council for Harmonization (ICH) in 1990, developing guidelines (S7B and E14) requiring preclinical assessment of human ether-à-go-go-related gene (hERG) channel inhibition and clinical evaluation of QT interval

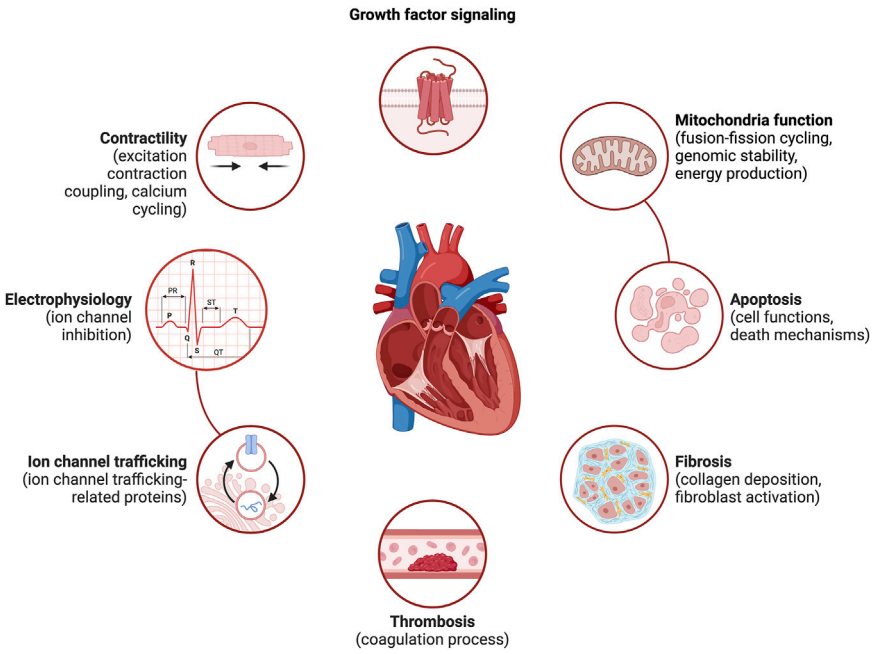


FIGURE 1
Overview of key pathophysiological mechanisms in cardiac dysfunction and cardiotoxicity. Schematic illustration of eight key mechanisms contributing to cardiac pathophysiology: growth factor signalling, mitochondrial function (fusion-fission cycling, genomic stability, energy production), apoptosis, fibrosis (collagen deposition, fibroblast activation), thrombosis (coagulation processes), ion channel trafficking, electrophysiology (ion channel inhibition), and contractility (excitation-contraction coupling, calcium cycling). These interconnected pathways play critical roles in chemotherapy-induced cardiotoxicity and other forms of cardiac injury. Created with BioRender.com

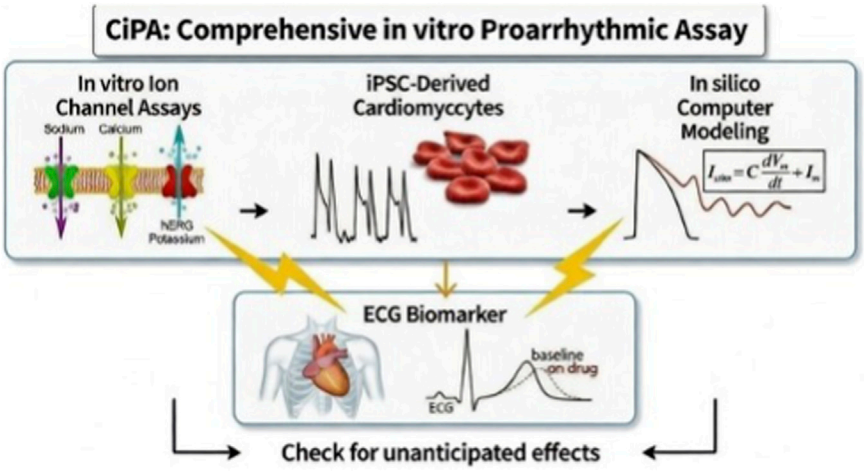


FIGURE 2
The Comprehensive *in vitro* proarrhythmia assay (CiPA) initiative workflow for cardiac safety assessment. The CiPA paradigm consists of four integrated components: (1) *in vitro* ion channel profiling to characterize multi-channel drug effects on cardiac ion currents (INa, ICaL, IKr, etc.); (2) human iPSC-derived cardiomyocyte (hiPSCs-CM) assays to validate integrated cellular electrophysiological responses; (3) *in silico* modeling to generate proarrhythmia risk scores from ion channel data; and (4) clinical ECG biomarker assessment to detect unanticipated effects in humans. This mechanistic approach improves upon the traditional hERG-QT paradigm by accounting for balanced multi-channel effects and reducing false-positive predictions. Inspired by the image in this publication. https://ascpt.onlinelibrary.wiley.com/doi/10.1002/cpt.896.

prolongation *in vitro* (European Medicines Agency, 2016). Although highly sensitive, these guidelines lack specificity, potentially, leading to unnecessary withdrawal of safe drugs due to sole reliance on QT prolongation and hERG channel blockage.

To address the limitations of ICH guidelines, the comprehensive *in vitro* Proarrhythmia Assay (CiPA) was introduced in 2013 (European Medicines Agency, 2016). CiPA employs an integrated approach involving: 1) *in vitro* evaluation of drug effects on seven critical human ventricular ion channels (INa, INaLate, ICaL, IKr, IKs, Ito, IK1), 2) *in silico* modelling of cardiac action potentials, 3) functional assessments using human induced pluripotent stem cell-derived cardiomyocytes (hiPSC-CMs), and 4) clinical validation via electrocardiographic (ECG) studies (Gintant et al., 2016). Functional readouts from patch clamp techniques, microelectrode arrays (MEA), voltage-sensitive dyes, and impedance measurements contribute data to computational models like the O'Hara-Rudy (Qauli et al., 2023; Romero et al., 2015; Park et al., 2019) ventricular model, enhancing predictive power regarding proarrhythmic risks. As mentioned before, cardiotoxicity can manifest itself not only as arrhythmia but also as contraction dysregulation and structural changes which calls for development of more holistic approaches to preclinically screen these modalities.

Despite these advancements, CiPA also exhibits certain limitations, including reliance on heterologous expression systems and immature phenotypes of hiPSC-CMs (Hwang et al., 2020). A schematic of the CiPA initiative is shown in Figure 2. Thus, the development and integration of more physiologically mature *in vitro* models, such as engineered cardiac tissues and 3D heart organoids, are critical to further refining CiPA's predictive capabilities (Dardano et al., 2024; Thomas et al., 2022). Notably, while CiPA provides a robust framework for assessing proarrhythmic risks of small molecules, it lacks standardized methods for evaluating drug-induced contractile dysfunction. MEA platforms—recommended by CiPA - offer powerful, label-free systems capable of simultaneously assessing electrophysiological activity and contractility. This makes them particularly promising for comprehensive cardiotoxicity evaluations—enabling monitoring electrical and mechanical alterations of cardiac models due to cancer treatments.

A notable limitation of *in vitro* assays such as MEA platforms is that it cannot capture the QT interval directly, which is an organ-level parameter. This is because QT interval as a parameter determined from ECG, is the result of superposition of the individual external field potentials of each single cardiomyocyte at the organ level. It is evident that monolayers of cardiomyocytes or 3D cell models do not resemble the full complexity of the human heart. This calls for methods that determine the cardiotoxic effects directly from the action potential. This forms the trend towards fabrication of 3D structures such as nanoneedles on microelectrode arrays or electroporation to achieve action potential like measurement which will be discussed in more depth in the section about Micro-electrode arrays (MEAs).

Prior to moving on to exploring MEA platforms as an *in vitro* assay to study cardiotoxicity, we need to understand how cancer therapies can cause arrhythmias and contractile dysfunction. Thus, establishing a foundational understanding of how action potentials (APs) are generated and how cardiac excitation-contraction coupling (ECC) functions physiologically is crucial.

Heart physiology: the excitation-contraction coupling journey

The heart's ability to effectively pump blood relies on the coordinated interplay between electrical activation and mechanical contraction of atrial and ventricular cardiomyocytes. Electrical impulses originate in the sinoatrial (SA) node, the heart's primary pacemaker, and propagate through the atria toward

atrioventricular node (AV). Following a brief conduction delay at the AV node, the impulse rapidly travels through the His-Purkinje network and finally reaches the ventricular cardiomyocytes. Central to this process is the cardiac AP, generated by sequential activation and inactivation of voltage-gated ion channels. The main voltage gated ion channels involved in this phenomenon are sodium, potassium and calcium channels (Grant, 2009). A complex interplay of signalling molecules and proteins and channels such as L-type calcium channels and calcium haemostasis in combination with T-Tubules couple action potential generation to cardiomyocyte contraction. Any changes in these channels can lead to improper electrical property and contractility of the cardiac muscle and cause abnormal heart function.

Cardiac action potential happens in five consecutive phases, spanning from phase 0 to phase 4. Each phase is intricately connected to the opening and closing of particular channels and the subsequent movement of associated ions. Phase 4 represents the resting membrane potential, which is maintained at around −90 mV by inward rectifier K^+ channels and Na^+/K^+-ATPase pumps. When electrical stimuli from conducting cells excite cardiac cells, the cell membrane potential becomes permeable to Na^+ ions, leading to rapid depolarization (phase 0). The influx of sodium typically lasts for about 1–2 milliseconds. After the action potential peak, there is a brief phase of rapid repolarization, known as phase 1, resulting from Na^+ channel inactivation and K^+ efflux from the cell. Phase 2, often referred to as the plateau, represents a balance between Ca^{2+} influx and K^+ efflux and is the longest phase. It is followed by phase 3, which is characterized by repolarization due to K^+ channel opening, resulting in K^+ efflux and the inactivation of Na^+ and Ca^{2+} channels. Ultimately, as the membrane becomes exclusively permeable to K^+ ions, the membrane potential returns to its resting state. Phase 2 is of short duration in atrial cardiomyocytes but prolonged in ventricular cardiomyocytes (Varró et al., 2021) and is critical for linking excitation to mechanical contraction via Excitation-contraction coupling (ECC).

Exquisite orchestration of action potential and contraction because of Excitation-contraction coupling contributes to continuous contraction and relaxation of cardiomyocytes and heart function. The underlying mechanism of EC coupling is attributed with changes of cytoplasmic calcium concentration $[(Ca^{2+})_i]$ (Figure 3). Ca^{2+} binds to troponin which results in sliding of the thick and thin sarcomeres, cell shortening, and development of pressure within the ventricles and pumping of blood. Intracellular calcium concentration increases during systole (contraction phase) and successful return of the intracellular calcium concentration to low enough levels during diastole (relaxation phase) to allow refilling of the heart with blood. As mentioned during the depolarization, L-type Ca channels open and consequently the slight increase in the intracellular Ca^{2+} concentration leads to large increase of Ca concentration in the dyadic space. Dyadic space is the region bounded by T-Tubule [invaginations of the surface sarcolemma occurring at each sarcomere (z-line)] and sarcoplasmic reticulum (SR) (responsible for calcium storage). The large increase of Ca^{2+} in this region leads to opening of SR Ca release channels [ryanodine receptors (RyR)] and release of Ca from the SR through a process called calcium-induced calcium release to the cytoplasm. Other proteins involved in EC coupling are NCX (Na^+/Ca^{2+} Exchanger) and SERCA (Sarcoplasmic/Endoplasmic Reticulum Ca^{2+}-ATPase) which are depicted in Figure 1 (Simple schematic depicting Action potential and contraction coupling). During relaxation of the heart, intracellular Ca^{2+} concentrations return back to cardiomyocytes' resting state and NCX and SERCA serve important roles in this phase (Crocini and Gotthardt, 2021; Izu et al., 2020; Dirksen et al., 2022; Ko et al., 2022; Liu Y. et al., 2023).

Alterations to the action potential and EC coupling lead to heart disease states. One alteration to the normal action potential of the heart is QT prolongation (Miller et al., 2014; Aronsen et al., 2013; Schram et al., 2002; Feher, 2012; Lei et al., 2018). QT prolongation primarily results from the inhibition of ventricular repolarization. Ventricular repolarization involves several potassium currents, with the primary component being the delayed rectifier potassium channel. From a molecular biology perspective, this channel is associated with the human-ether-a-go-go related gene (hERG) (Elming et al., 2002). The duration of action potentials in ventricular cardiomyocytes is closely linked to the QT interval observed on an electrocardiogram (ECG). The length of the QT interval, however, is primarily regulated by an equilibrium between the inward late sodium current (INaL), which tends to prolongation of the action potential duration (APD), and the outward potassium currents, which tend to shorten APD (Biet et al., 2015; Kannankeril and Roden, 2007). Prolongation or shortening of the QT interval has been linked to ventricular tachyarrhythmias, which increase the risk of sudden cardiac death. These conditions can either be congenital, resulting from ion channel mutations, or acquired due to other factors.

Prolonged QT interval can also be a trigger for dangerous ventricular tachyarrhythmias like torsade de pointes (Kannankeril and Roden, 2007). It is important to note that not all drugs that lengthen the QT interval necessarily raise the risk of torsade de pointes. Other parameters such as ectopic beats, dispersion of ventricular repolarizations and more specially EADs serve as better prognostic factors of TdP (Belardinelli et al., 2003; Little et al., 1990). QT prolongation, on its own, serves as a somewhat dependable indicator of the likelihood of experiencing torsade de pointes (Elming et al., 2002; Frothingham, 2001). TdP is an infrequent polymorphic ventricular tachyarrhythmia characterized by a distinctive twisting of the QRS complex around the baseline, which sometimes resolves on its own but can also lead to genuine ventricular tachycardia, ventricular fibrillation, and sudden cardiac death (El-Sherif et al., 1997).

The most common serious cardiac arrhythmia is atrial fibrillation (AF). AF occurs when abnormal APs start firing in the atria and can override the heart´s natural pacemakers. This way the SAN, AVN and ventricular pacemakers can no longer control the heart's rhythm. Consequently, AF causes irregular and often abnormally fast contraction of the atrial cardiomyocytes. This leads to various symptoms such as irregular heart rate, shortness of breath and tiredness. In worst cases, AF can lead to heart failure (HF), stroke and sudden cardiac death (Hajjar et al., 2021; Brundel et al., 2022; Menichelli et al., 2021; Qin et al., 2024).

Heart failure is the inability of the heart to meet the body's metabolic demands. It is the primary endpoint of many cardiac pathologies and the secondary health complication of diseases such as diabetes and can occur due to cancer treatments. Disruption of calcium handling in cardiomyocytes can disturb EC coupling and is

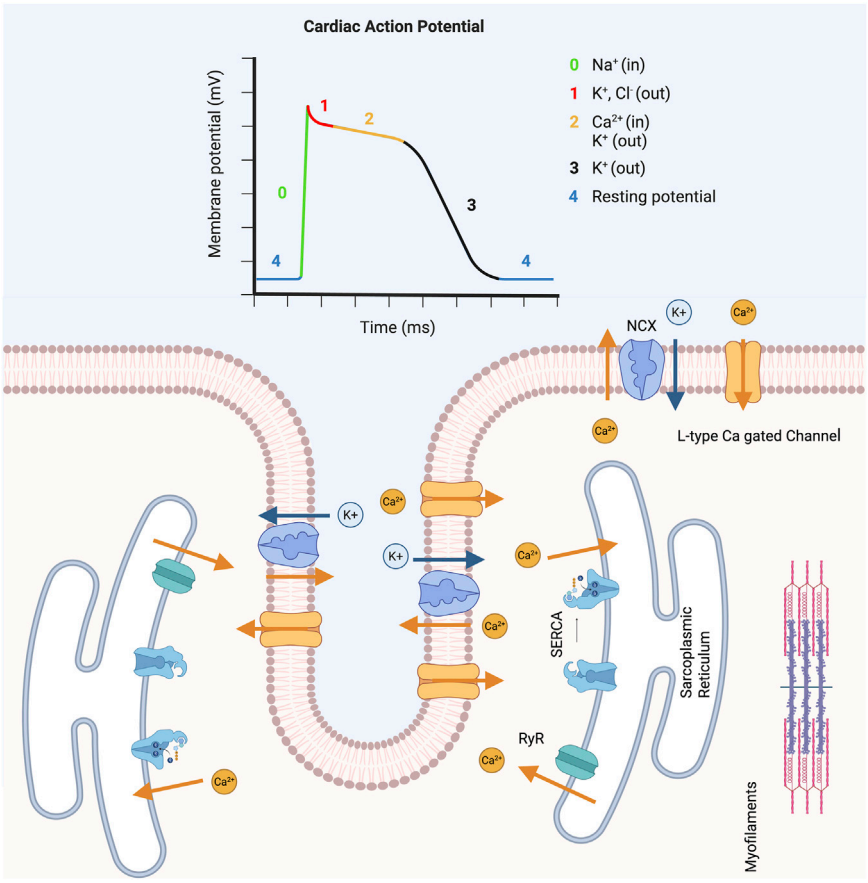


FIGURE 3
Cardiac action potential phases and molecular mechanisms of excitation-contraction coupling. The cardiac action potential (top) proceeds through five phases: Phase 4 (resting potential, ~-90 mV; IK1, Na^+/K^+-ATPase); Phase 0 (rapid depolarization; INa); Phase 1 (early repolarization; Ito); Phase 2 (plateau; ICaL vs. IKr/IKs balance); and Phase 3 (final repolarization; IKr/IKs). Excitation-contraction coupling (bottom) is initiated when L-type Ca^{2+} channels open during Phase 2, allowing Ca^{2+} entry into the dyadic space between T-tubules and sarcoplasmic reticulum (SR). This Ca^{2+} influx triggers SR Ca^{2+} release via ryanodine receptors (RyR), increasing cytoplasmic [Ca^{2+}]- and enabling troponin-mediated myofilament contraction. Relaxation occurs through Ca^{2+} reuptake by SERCA into the SR and Ca^{2+} extrusion via NCX. This precise Ca^{2+} cycling orchestrates the systole-diastole cycle essential for cardiac pump function. Created with BioRender.com

a prominent factor in arrhythmia (such as AF) and HF. Dysregulation in calcium handling may lead to abnormal cardiomyocytes contractility and pathological remodelling. The mentioned alterations to action potential and EC coupling arise because of various cellular and molecular mechanisms (Swain et al., 2003; Hahn et al., 2021; Kansakar et al., 2021).

Mechanisms of cancer treatment induced cardiotoxicity (CIC)

In cancer treatments and more specifically chemotherapy, chemotherapy agents affect all major organs and one of the most prevalent adverse effects is cardiotoxicity (Hahn et al., 2014). Regarding immunotherapies, due to the targeted nature of the therapy, cardiotoxicity is not as pronounced. Consequently, a new medical field known as cardio-oncology has emerged in recent years to address these concerns (Albini et al., 2010; Bellinger et al., 2015). One notable adverse effect of chemotherapy-induced cardiotoxicity is arrhythmia, primarily characterized by QT prolongation. Prolonged QT intervals indicate longer action potential durations and delayed ventricular repolarizations in cardiac myocytes, often resulting in ventricular tachyarrhythmias (Elming et al., 2002). The AP and ECG signal changes in response to chemotherapeutics and immunotherapeutic treatments (Table 1). Another notable adverse effect of CIC is contraction dysregulation which is a contributing factor of heart failure. Modulation of cardiac contractility has been proposed as an effective treatment strategy for heart failure. Contraction dysregulation can occur as the result of excitation-contraction coupling defects (Kistamás et al., 2020).

Here, we briefly review the mechanisms of cardiotoxicity induced by a family of chemotherapeutics called anthracyclines (such as Doxorubicin). Anthracyclines cause irreversible cardiac damage primarily through oxidative stress on cardiac myocytes and mitochondrial dysfunction which alters the morphology of cardiomyocytes. For an in-depth review of chemotherapeutics induced cardiotoxicity and their mechanism of action please refer to the review by (Adhikari et al., 2021).

TABLE 1 List of cancer treatments with their associated cardiotoxic effect.

Drug	Mechanism of cardiotoxicity	Effect on action potential morphology	References
Anthracyclines (especially DOX)	Reactive oxygen species, mitochondrial dysfunction, cardiolipin interaction and accumulation of DOX within the mitochondria, interaction with ferric ions and generation of free radicals.	prolongation of the QT and possible reduction in the dV/dt_{max}	Benjanuwattra et al. (2020)
Alkylating agents (especially cyclophosphamide)	Reactive oxygen species and mitochondrial dysfunction, induced apoptosis through the expression of BAX located in the mitochondrial membrane.	QTc prolongation, T-wave and ST-segment changes	Hahn et al. (2014)
Antimetabolites (especially 5-Fluorouracil)	Cellular damage and induction of ischemia.	QTc prolongation and increased AT dispersion, ST-T wave changes	Hahn et al. (2014)
Histone deacetylase inhibitors (Vorinosat)	Transcriptional changes of genes required for ion channel trafficking and localization to the sarcolemma.	QTc prolongation of >500 ms (severe arrhythmias and EKG changes).	Hahn et al. (2014)
Immunomodulatory drugs (Thalidomide, Lenalidomide)	In some cases, reactive oxygen species and mitochondrial damage.	Dose dependent QTc prolongation of Thalidomide No QTc prolongation of Lenalidomide	Ha et al. (2004), Chen et al. (2013)
Platinum compounds (Cisplatin)	Mainly reactive oxygen species and subsequent induction of oxidative stress that change into prothrombotic condition.	Incidence of QTc prolongation with Cisplatin has not been reported.	Porta-Sánchez et al. (2017)
Targeted cancer therapy	Various drugs exist in this category with different mechanisms of action. Please refer to the following references.	QT interval prolongation associated with TdP	Hahn et al. (2021), Liu Y. et al. (2023)
Tyrosine kinase inhibitors (Alectinib, Cretinib, Crizotinib, Dasatinib, Nilotinib)	Based on target effects and off target effects via inhibition of several other kinases important in the maintenance of cardiovascular function.	QTc prolongation with Nilotinib, Crizotinib, Alectinib and Certinib having QTc prolongation of more than 500 ms.	Hahn et al. (2014)
Human epidermal growth factor receptor 2	Reduced ErbB2 signaling from the cell membrane due to binging of trastuzumab to the extracellular domain of ErbB2/HER2 (ErbB2 family of receptors have cardio-protection effects), blocks the proteolytic cleavage of ErbB2, antibody dependent cell mediated cytotoxicity and degradation of the ErbB2 receptor.	QTc prolongation of > 500 ms	Hahn et al. (2014)

It is worth noting that anthracyclines, like doxorubicin, are included in more than half of the anticancer therapies utilized today. Doxorubicin (DOX) is an anthracycline antibiotic that was first employed as a cytotoxic drug in 1969. It has demonstrated significant efficacy in treating various cancer types, including solid tumors, breast cancer, lung cancer, soft tissue sarcoma, and various metastatic malignancies (Tacar et al., 2013). Nonetheless, its use is constrained due to a substantial incidence of side effects, including hematopoietic system suppression (Swain et al., 2003). One reason for DOX induced cardiotoxicity is oxidative stress which leads to degeneration of cardiomyocytes (Songbo et al., 2019). Another reason is the interaction of DOX with cardiolipin (an important component of the inner membrane of mitochondria) which yields irreversible binding of these two components and accumulation of DOX in the mitochondria. DOX interacts with ferric ions (Fe 3+) as well and this complex interacts with the cell membrane, causing lipid peroxidation and generating free radicals. Additionally, DOX leads to apoptosis of cardiac muscle cells through the intrinsic and extrinsic pathways (Sheibani et al., 2022). Anthracycline (ATN) induced cardiotoxicity is significantly dose dependent (Sawicki et al., 2021). For instance, at cumulative doses ranging from 300 to just under 550 mg/m^2, approximately 2.2% of patients receiving DOX experience heart failure (HF).

This risk substantially increases to 26%–30% when total doses fall within the range of 500–800 mg/m^2. Overall, the primary mechanisms contributing to cardiac damage caused by doxorubicin include the inhibition of topoisomerase 2 beta (TOP2) and the generation of reactive oxygen species (ROS) (Sheibani et al., 2022; Kong et al., 2022; Nousiainen et al., 1999). In doxorubicin cardiotoxicity, progression of EC coupling defects is evident and not accompanied with cellular hypertrophy or AP prolongation at early stages. The cardiac myocytes unloaded contraction reduced after 15 weeks

TABLE 2 List of CiPA recommended *in vitro* assays for screening cardiotoxicity induced by pharmaceuticals.

Name of the technique	Endpoint features	Advantages	Disadvantages	Review articles
Microelectrode arrays	Indirect features extracted from APs: ● Repolarization ● Depolarization ● Beat timing ● Arrhythmia occurrence ● Signal propagation	● High throughput ● Screens for all ionic channels ● Non invasive	● Indirect measurement of APs ● Can be used with heart organoids	Tanwar et al. (2022), Ferguson et al. (2019)
Automated patch clamp	● Different steps of Action potential signal: All phases. ● Arrhythmia occurrence	● Direct measurement of intracellular action potential ● High throughput	● Limited cell types ● Invasive ● Cannot be used with heart organoids	Hamill et al. (1981), Obergrussberger et al. (2021), Seibertz et al. (2022)
Voltage imaging (chemical indicators)	● Different steps of Action potential signal: All phases. ● Arrhythmia occurrence	● Medium throughput ● Direct measurement of intracellular action potential	● Bleaching ● Toxic to cells ● Only short time monitoring of cells APs ● Special equipment is needed for imaging such as fast camera	Vogt (2019), Knöpfel and Song (2019)
Voltage imaging (Genetically encoded voltage indicators)	● Different steps of Action potential signal: All phases. ● Arrhythmia occurrence	● No photo bleaching ● Long term monitoring of cells ● Medium throughput	● Special equipment is needed for imaging such as fast and camera	Vogt (2019), Knöpfel and Song (2019)
Impedance	● Cardiomyocyte contractility ● EAD occurrence.	● Non-invasive ● chronic measurement ● High-throughput	● Only recording EAD ● Not recording AP or FP	

of doxorubicin treatment *in vivo* which occurs in models of HF accompanied by cardiac remodelling too (Biet et al., 2015).

Cardiotoxicity due to DOX administration is one of the most studied chemotherapeutic so far. However, even for doxorubicin, the information of action potential and field potential changes is considerably scarce. In regards to while there is limited available information on the impact of DOX on cardiac arrhythmias *in vitro*, several reports have discussed its arrhythmogenic potential in cancer patients clinically (Lenneman and Sawyer, 2016). Overall, treatment with DOX has the potential to result in various cardiac issues, including dysfunction of the SA node, slower conduction of electrical signals within the ventricles, heightened susceptibility to arrhythmias, and abnormalities in ventricular repolarization (Swain et al., 2003).

Cancer patients treated with DOX frequently experience the occurrence of supraventricular and ventricular extrasystoles, which are abnormal heart rhythms. Extrasystoles, also referred to as triggered activities, arise from phenomena known as early afterdepolarizations (EADs) or delayed afterdepolarizations (DADs), which become significant enough to initiate action potentials. Furthermore, DOX has been found to notably prolong the QT interval in various studies (Nousiainen et al., 1999). When DOX is administered alongside an Ikr blocker like erythromycin, it can exacerbate QT interval prolongation, potentially increasing the risk of developing torsade de pointes, a specific type of ventricular tachycardia. However, it is important to note that DOX does not affect the resting membrane potential of cardiac cells (Jensen et al., 1984). The impact of DOX on maximal action potential upstroke velocity (dV/dtmax) and AP amplitude (APA) appears to be controversial. Despite the limited reports and the need for further research, these findings suggest that the consequences of DOX administration on the morphology of action potential are prolongation of the QT and possible reduction in the dV/dtmax (Benjanuwattra et al., 2020).

The various adverse effects of DOX on the heart and the effect of DOX administration on cardiac action potential and ionic currents have been reviewed elsewhere in more detail (Benjanuwattra et al., 2020). DOX can cause TdP (as it caused the death of one hypokalaemic patient treated with DOX) and dexrazoxane can prevent the proarrhythmic mechanism of acute DOX treatment (Ducroq et al., 2010). It would be interesting to scrutinize the AP and FP recordings of *in vitro* models of DOX (2D and 3D) and study the underlying mechanisms of arrhythmia induced by DOX. Scrutinizing this phenomenon can help develop less cardiotoxic combination therapy of DOX (Pereira-Oliveira et al., 2019). This will help lift the constraint on using this chemotherapeutic in cancer patients especially in the paediatric population. 50,000 of paediatric patients out of 500,000 cancer survivors pass away due to doxorubicin induced cardiotoxicity (Lei et al., 2018).

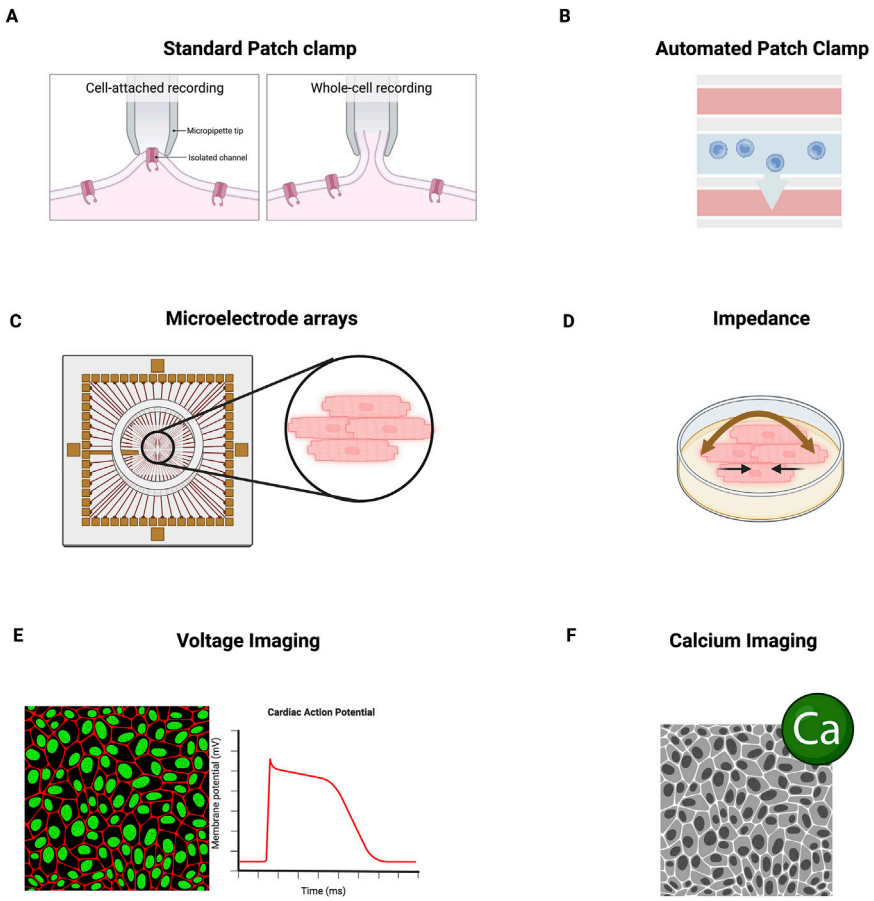


FIGURE 4
Overview on the CIPA-recommended in-vitro assays for cardiotoxicity screening, enabling comprehensive cardiotoxicity evaluation of cancer therapeutics: **(A)** Conventional patch clamp and **(B)** automates patch clamp for transmembrane action potential recording using; **(C)** MEAs for for extracellular field potential monitoring and **(D)** impedance based contractility assessment; **(E)** voltage imaging (green fluorescence with voltage trace) and **(F)** calcium imaging (Ca2+ transients, pseudo-coloured) for excitation-contraction coupling analysis. Created with BioRender.com

Cancer treatment-induced cardiotoxicity assessment (CIC) *in vitro*

The incidence of cancer rises significantly with advancing age, particularly within the age range of 40–80 years, this is concomitant with the increased risk of cardiac diseases. With the advancement in cancer chemotherapy, the life expectancy and survival rate of cancer patients has increased. But the substantial adverse effects associated with the chemotherapeutics remain a significant challenge (Lenneman and Sawyer, 2016). CIC or cancer treatment induced cardiotoxicity can be detected preclinically *in vitro* (Gintant et al., 2019). For chemotherapeutics such as small molecules this can fall under the CiPA regulations and its recommended *in vitro* assays (see Table 2) discussed in the introduction.

Based on CiPA there are various approaches that evaluate the electrophysiological markers of proarrhythmia using hSC-CMs and in some cases heterologous expression of ion channels *in vitro* (Colatsky et al., 2016). These approaches include transmembrane action potentials, MEA recordings, impedance measurements, voltage sensitive dyes and calcium transients (Figure 4).

Techniques for transmembrane action potential measurement are patch clamp and automated patch clamp. Patch clamp is the gold standard *in vitro* method in electrophysiological recordings of electroactive cells (Stoelzle et al., 2011). Patch clamp technique helped in understanding the underlying mechanisms of action potential (Verkhratsky and Parpura, 2014). In 1991 Neher and Sakmann won the Nobel Prize in physiology or medicine for their pioneering work in development of patch clamp. Patch clamp functions by forming a tight seal (giga-ohm seal) between a finely polished glass pipette and cell membrane, enabling precise recordings of ionic currents and membrane potentials. The conventional patch clamp is low throughput and is technically demanding. Therefore, it is unsuitable for high throughput screening (HTS). Automated patch clamp on the other hand have worked on becoming high throughput (up to 384 simultaneous recordings) and can be used with standard cell lines such as HEK and CHO with heterologous expression of cardiac ion channels and hSC-CMs with fully integrated ion channels (Gintant et al., 2019; Stoelzle et al., 2011; Kramer et al., 2020).

In the late 1990s the first attempts to automate voltage clamp recordings started with the NeuroPatch system from Sophion

Bioscience (Asmild et al., 2003). It had the capability of eight parallel recordings. But due to technical issues it did not gain commercial success. With the introduction of planar array electrophysiology, the embedding of an array of micro apertures in a planar substrate with 1–2 microns openings, the major limitations of automated patch clamp could be resolved. Cells in suspension are positioned onto apertures to form a high resistance seal using suction or gravity (Stoelzle et al., 2011). The first commercial APC systems based on planar electrophysiology were IonWorks HT and Quattro System. In 2006, Nanion launched Patchliner, a fully automated planar patch clamp that could record from two up to six wells simultaneously with giga ohm seals (Annecchino and Schultz, 2018; Zheng et al., 2004). The key features of APC technology are microfluidics incorporated in it, temperature control, current clamp capability (clamping the membrane current and recording membrane voltage) and unattended operation (Kodandaramaiah et al., 2012). However, the APCs lead to dialysis of the cells and are not amenable for chronic measurement of drug effects and cannot be used readily with multicellular organoids which are emerging as novel, more advanced cardiac models. But the advantage of transmembrane action potential techniques are the most easily comparable obtained results with *in silico* models and ECG effects. APCs have been used in validation studies for CiPA with heterologous standard cell lines and hSC-CMs (Vicente et al., 2018).

MEAs alongside hSC-CMs are one of examples of developing assays to give an integrated assessment of multiple ion channels contributing to the cardiac action potential (AP) (Millard et al., 2016). MEA technology is used to study electroactive cells for decades in a label-free manner and can noninvasively interface with established cardiomyocyte networks. This way, we can get mechanistically functional measures of the cardiac AP without perturbing the network long-term. The field potentials recorded from MEAs correlates with the cellular action potential duration (APD) and with the QT interval of ECG; as all three are governed largely by the same repolarizing currents. FP measures the duration of cardiomyocytes repolarization. It is important to note that however, that FPD and APD are not equivalent measurements. The FP is an extracellular signal at the tissue level, whereas the APD reflects the transmembrane potential of an individual cell. As a result of the nature of FP, the FP signal is influenced by a variety of factors that do not reflect the intracellularly measured APD. These factors could be electrode geometry, cell-electrode coupling, cell density and degree of gap junctional coupling and conduction properties in the case of 2D monolayer of cardiomyocytes. Typical end points for MEA studies include field potential duration and incidence of EADs (Early after depolarizations), beat frequency, amplitude of initial depolarizing spike (a measure of sodium current) and conduction patterns.

In a study, conventional and composite biomarkers from MEAs were analysed for running an *in silico* model to validate and classify drug induced channel block (Tixier et al., 2018). The composite markers were area under the curve of repolarization wave, the repolarization centre, the repolarization width, the FP notch and the depolarization width which were comparably better at classifying drugs in causing TdP. In a technical paper by Millard et al. (2016), the methodology for MEA recording of hSC-CMs is extensively described. There are several studies for optimizing and validating MEA assay for CiPA (Blinova et al., 2017; Millard et al., 2018). There is a cross-site blinded CiPA Pilot study carried out in 2018, eight drugs for validation of MEA and hSC-CMs which was then followed by another study that two hSC-CM cell lines were chosen. This study characterized the site-to-site variability of assessment of FP measurement using microelectrode arrays. The control drugs and the test drugs (28 drugs in total) were added sequentially, and voltage sensing dyes (VSOs) were also used to check for variabilities in AP evaluation (Blinova et al., 2017). A study was also carried out to evaluate the electrophysiology response of the hSC-CMs with MEAs and VSOs on 20 drugs. This was done to evaluate the possibility of inclusion of these techniques and hSC-CMs for inclusion in the CiPA screening platform.

Impedance has also been used with hSC-CMs to quantify modulators of cardiac contractility (Pfeiffer-Kaushik et al., 2019). Impedance measurement of contractility can be carried out with MEAs and other specifically designed sensors. In application notes from Axion biosystems, MEAs with capabilities for both field potential and impedance measurement were used. In these MEAs both contractility and arrhythmia in hSC-CMs can be evaluated (Streeter et al., 2026; Axion Biosystems, 2026a; Axion Biosystems, 2026b). In another instance, uniquely designed interdigitated electrodes were used for impedance measurement. The xCELLigence Cardio system uses impedance technology based on interdigitated electrode configurations in a 96 well format and was used with 49 compounds in a concentration response mode to study the potency for modulation of CM beating which is a surrogate biomarker for contractility (Xi et al., 2011). There is also the possibility of using a pair of electrodes or a four-point measurement setup for measuring impedance. Impedance can also be used to detect EADs along with cardiac contractility (Xi et al., 2011).

Voltage and calcium sensitive probes are recently developed *in vitro* assays proposed by CiPA for cardiotoxicity assessment. In the twenty-first century, optical fluorescent probes emerged in the 1980s and provide spatial resolution down to a few cells or cellular organelles. Calcium and voltage indicators can pinpoint transient calcium fluctuations inside the cells and changes in the duration and time course of repolarization (providing measures of action potential triangulation) respectively (Chemla and Chavane, 2010). Calcium is a common secondary messenger known to regulate EC coupling in cardiomyocytes. The first used calcium sensitive dyes (e.g., fura-2) were not cell membrane permeable; in next-generation of calcium sensitive dyes, they became cell membrane permeable by adding an ester group (e.g., fura-2 AM) An associated problem with dyes is the blindly staining of different cells with different calcium and voltage signalling properties in one preparation. Genetically encoded calcium indicators (GECIs) – introduced in 2007- solve this issue and provide additional advantages such as reproducible preparations, photo-bleaching and avoidance of potentially harmful staining procedures (Mutoh et al., 2012; Mollinedo-Gajate et al., 2019).

In the case of voltage sensing, genetically encoded voltage indicators (GEVIs) were first published in 2007. Since then, the fast-paced progress in GEVIs has made them outperform organic voltage-sensitive dyes (VSDs) (Hortigon-Vinagre et al., 2021). In a recent, "Best Practices" paper on the use of hSC-CMs for preclinical cardiotoxicity screening, cardiac optical mapping techniques such as calcium and voltage sensitive dyes, or GECIs and GEVIs were used

alongside MEAs (Gintant et al., 2020). Recently, a cardiac electrophysiology plate-reader was developed by Allan et al. (2023) for high-resolution and high-throughput assessment of hSC-CM (Blinova et al., 2017; Blinova et al., 2018) electrophysiology using VSDs, calcium sensitive dyes and GECIs. MEAs and VSOs were used in international multi-site study for pro-arrhythmia assessment.

Despite these technological advancements, the integration of CiPA assays for screening cancer therapy-induced cardiotoxicities remains limited, particularly concerning immunotherapies and advanced cell-based treatments. Precision medicine approaches utilizing these assays, notably MEAs and advanced optical methods, show substantial promise in enhancing cardiotoxicity assessments, enabling the development of safer, more targeted cancer therapies tailored to individual cardiovascular risk profiles.

One of the most promising *in vitro* tools recommended by the CIPA initiative are microelectrode arrays (MEAs) which will be discussed in the next section.

Micro-electrode arrays (MEA)

Overview of MEAs and their emergence for cardiotoxicity assessment

One of the most promising *in vitro* tools recommended by the CiPA initiative are microelectrode arrays (MEAs). MEAs are (usually) planar chip substrates with microelectrodes on top that allow for bioelectrical and electrophysiological measurement. Hence, MEAs interface the electronics with the biological cells, enabling real-time, long-term, label-free and continuous readout. There are three primary electrical operation principles MEAs, which are explained in the following passages: (1) voltage recording from electrogenic biological cells such as cardiomyocytes, (2) impedance measurement of biological matter, e.g., for contraction recording of cells, and (3) for electrical stimulation of cells, e.g., cardiomyocyte by applying in voltages but also electroporation to gain a permeable. Depending on the electrode design and materials, all three measurement modes can be implemented on a single chip, referred to as a multimodal MEA (Schmidt et al., 2024). Furthermore, MEAs are available for *in vitro* and *in vivo* investigations, but the further text focuses on the *in vitro* use. At first, MEAs emerged for voltage measurements more than 50 years ago, following the pioneering work by Thomas et al. (1972). In this initial work, cardiac cells were utilized by seeding dispersed embryonic chick heart tissue on a MEA. This led to the formation of a confluent and contracting cell layer on top of the electrodes. With every contraction of the cardiomyocyte, an orchestrated opening and closing of ion channels results in changes of the transmembrane voltage, known as the action potential (AP), which is typically recorded using patch clamp, but can also be recorded using MEAs, as shown later. The electrical ion currents through the membrane during the AP also affect the electrical potential changes in the vicinity of the cells, called the field potential, which is typically recorded by external electrodes like MEAs. Using the MEA to measure the field potential of chick heart cells, Thomas et al. successfully recorded characteristic triphasic voltage signals with a peak-to-peak amplitude of 20–2,500 μV induced by the cardiac cells.

These first MEAs were purely passive MEAs without any signal processing integrated directly on the MEA. For the readout, MEAs are typically placed into a head stage that interfaces the readout device with contact pads on the MEA, usually using pogo pins. In passive MEAs, the electrodes are directly interconnected to these contact pads, and all signal processing happens in the readout device. Since then, the MEA technology has evolved, and besides the still-used passive MEAs, active MEAs with signal processing, e.g., amplification directly integrated into the MEA, have been developed. Active MEAs typically use Complementary Metal-Oxide-Semiconductor (CMOS) logic and are therefore referred to as CMOS-MEA. Andreas Hierlemann's research group from ETH Zurich published work in 2010 on developing a high-density MEA chip based on CMOS technology (Frey et al., 2010; Heer et al., 2004). The overall chip size was $7.5 \times 6.1\,\mathrm{mm}^2$ with a MEA size of $2 \times 1.75\,\mathrm{mm}^2$. More than 10,000 metal electrodes were distributed in this area of 3.5 square millimetres. The diameter of the microelectrode is 7 μm, with a spacing of 18 μm. The chip was also equipped with a signal excitation and signal acquisition module, which can realize two-way communication with myocardial and electrogenic cells. CMOS-MEAs offer the ability for on-chip amplification and further signal processing, leading to increased SNR and the possibility of a higher electrode density, with up to 236,880 electrodes at single-neuron resolution (Suzuki et al., 2023).

Passive MEAs lack integrated electronics that rely on expensive, stiff, non-transparent semiconductors, and are therefore more flexible in terms of the substrate material. Consequently, passive MEAs can be transparent for parallel optical assays, e.g., using glass (Connolly et al., 1990), or flexible and low-cost using plastic substrates and standard liquid crystal display technology (Charkhkar et al., 2012). The electrodes can be fabricated with different materials to match requirements, such as high conductivity (e.g., platinum), chemical surface modifiability (e.g., gold via thiol modification), and optical transparency (e.g., Indium Tin Oxide, ITO). Furthermore, the electrode surface can be coated with materials like Titanium nitride (Janders et al., 1996) or advanced materials like PEDOT:PSS-MWCN nanocomposites (Liang et al., 2025) to lower the electrode impedance.

MEA fabrication is typically performed using microfabrication technology such as photolithography, sputtering, vapor deposition, and reactive ion etching. Besides these standard processes, new approaches such as inkjet printing of the electrodes and isolation are emerging (Bachmann et al., 2017). In the fabrication of MEAs to prevent a short circuit between the interconnects due to the ion-containing, thus conductive, cell culture medium, the interconnects of both passive and active MEAs need to be isolated. Isolation is typically performed using polymers such as SU-8, which features a lower electrical permittivity but worse optical properties, Parylene, or ceramics such as silicon nitride (Schmidt et al., 2021).

For verifying electrical signals using established gold standard optical methods, simultaneous optical and electrical readout is often crucial. Therefore, new and novel works, e.g., by Hu et al. (2025), are utilizing transparent thin-film transistor technology. These TFT-MEAs for the simultaneous recording of optical, voltage, and impedance signals from cardiomyocytes, featuring

TABLE 3 A comparative analysis of the principal microelectrode array (MEA) architectures for the assessment of cardiotoxicity is presented herein. The table presents a comparison of passive, complementary metal-oxide-semiconductor (CMOS), and thin-film transistor (TFT) MEAs based on their structural, optical, and electrical performance parameters (Thomas et al., 1972; Hu et al., 2026).

Parameter	Passive MEA	CMOS MEA (active)	TFT MEA
Electrode Count and Pitch	Low to moderate (~60–250); large pitch (>100 μm)	Ultra-high (>200,000); fine pitch (<20 μm)	High (10,000–50,000); moderate pitch (<50 μm)
On-Chip Processing	None (passive routing; requires external amplification)	Full integration (on-chip amplification, filtering, multiplexing)	Basic integrated switching (row/column matrix addressing)
Signal-to-Noise Ratio (SNR)	Moderate (susceptible to electromagnetic interference and parasitic cable capacitance)	High (minimized cable noise via localized, pixel-level amplification)	Moderate to high (active matrix addressing minimizes cross-talk)
Optical Transparency	High (when using glass/polymer substrates and ITO electrodes)	Opaque (monolithic silicon substrate)	High (fabricated on glass/quartz with ITO electrodes)
Substrate Flexibility	Excellent (compatible with polymers, glass, and elastomers)	Rigid (restricted to bulk crystalline silicon wafers)	Good (adaptable to flexible thin-film polyimide/plastics)
Cardiotoxicity Screening Merit	Cost-effective; seamless integration with high-resolution inverted optical microscopy	Sub-cellular spatial resolution; maps micro-conduction velocity propagation	Enables simultaneous high-density electrical mapping and optical pacing/imaging
Primary Limitation	Poor spatial resolution; cannot resolve micro-conduction block or localized arrhythmias	High fabrication cost; precludes transmission light microscopy/optogenetic trans-illumination	Higher intrinsic electronic noise than CMOS; complex thin-film manufacturing

22,500 transparent ITO electrodes. The individual benefits of passive (transparent, low price) and active (high electrode density, increased SNR) MEAs have led to the co-existence of both systems nowadays (Hu et al., 2025). To systematically evaluate these architectural choices for cardiotoxicity screening, it is necessary to carefully balance the physical, optical and electrical properties of passive, CMOS and TFT MEA platforms. While passive systems prioritise substrate flexibility and cost-effectiveness, CMOS-MEAs demonstrate superiority in terms of spatial resolution, and TFT-MEAs offer a compromise by facilitating high-density recording without compromising optical clarity. A detailed comparative classification of these three principal architectures is outlined in Table 3.

MEAs enable non-invasive and medium-throughput acquisition of extracellular field potentials (FPs) from multiple sites when interfaced with electroactive cells such as cardiac cells. A broad range of electrogenic cell types, including chicken ventricular myocytes (Tanwar et al., 2022), iPSC-derived cardiac embryoid bodies (Ferguson et al., 2019), single-chambered heart organoids (Hamill et al., 1981), and finally multi-chambered heart organoids (Obergrussberger et al., 2021), were investigated by MEA voltage recording.

Secondly, in combination with a voltage or current source, MEAs enable non-invasive impedance measurements of all types of cells–electroactive and non-electroactive cells- and the surrounding medium. The impedance changes due to the cardiomyocyte shape and position change during contraction were used to assess the contractility of cardiomyocytes (Seibertz et al., 2022) on a MEA. A work of Wang et al. (2020) showed the use of impedance measurements for contractile stress measurements with a relative error of 3.73% compared to AFM measurements. In another study, the xCELLigence Cardio system, which utilizes impedance technology in a 96-well format, was used with 49 compounds in concentration-response mode to investigate the potency of modulation of CM beating, a surrogate biomarker for contractility. There is also the possibility of detecting EAD by impedance (Zheng et al., 2004). In application notes from Axion Biosystems, MEAs with capabilities for both field potential and impedance measurement were used. In these MEAs, both contractility and arrhythmia in hiPSC-CMs can be evaluated (Axion Biosystems, 2026a).

Finally, a voltage source can also be applied between two electrodes to electrically stimulate or electroporate cells, such as cardiomyocytes. In the case of stimulation, it can be used to functionally mature cardiomyocytes, leading to a 100.75% higher field potential amplitude and a 54.89% reduction in the beating period (Liang et al., 2025). Electroporation on the other hand can be used to locally permeabilize the membrane, allowing for the direct recording of action potentials with minimal effect on cell viability, a technique referred to as open-cell recording. Braeken et al. (2012) demonstrated that open-cell measurements are feasible with minimal impact on cell viability and electrical behaviour. Furthermore, they showed the importance of open-cell recording for the screening of cardiotoxicity in drugs, using the example of dofetilide, which prolongs the action potential. The effect was only visible using the open-cell recording but not the extracellular field potential measurement. The same principle was also successfully used by Jans et al. (2017) to measure the effect of the drug quinidine in human induced pluripotent stem cells (hiPSCs) and verapamil in neonatal rat cardiomyocytes. While on-chip microelectrodes enable highly localized electrical pacing and electroporation, their small geometric surface area introduces distinct practical constraints during repetitive or long-term stimulation protocols. Managing the hardware parameters and electrochemical boundaries such as avoiding hydrolysis is critical. Under repeated, high-frequency pacing, the high current densities required at the micro-interface can easily exceed the material's safe charge injection capacity (CIC). This triggers irreversible faradaic reactions, culminating in electrode degradation, a progressive increase in electrode impedance, and local gas bubble generation via electrolysis. Consequently, advanced

materials with a high CIC are required if on-chip microelectrodes are used for stimulation. In long-term cardiac stimulation studies, macroscopic external graphite or platinum electrodes are often preferred to ensure stable and reproducible global pacing (Hu et al., 2026; Tandon et al., 2009; Sun et al., 2025). With the emergence of heart organoids or cardioids, which are 3D biological models resembling early stages of heart development, the need for combining flexible and unconventional electronics with conventional MEAs has emerged. As such, the development of 3D MEAs based on Kirigami design (Yang et al., 2024), Stretchable mesh MEAs (Le Floch et al., 2022), planar mesh MEAs (McDonald et al., 2023), self-folding shell MEAs (Huang et al., 2022), and buckled MEAs (Park et al., 2021). These constructs have been used to study brain organoids and assembloids. However, they can also be used for the electrophysiological screening of heart organoids. MEAs have been commercialized, and their products are primarily manufactured by MultiChannel Systems (Germany), Ayanda Biosystems (United States), Panasonic (Japan), MaxWell Biosystems AG (Switzerland), and 3Brain (Switzerland). Collectively, these developments have positioned MEAs as essential tools in cardiotoxicity screening, bridging electrophysiological insights with advanced cardiac models, including iPSC-derived cardiomyocytes and heart organoids. Moreover, integration of impedance measurement and stimulation capabilities allows simultaneous electrophysiological and mechanical assessments, maturation, and local electroporation on a single device. In the next chapter, we will examine the theories underlying field potential recording and impedance measurement (Suk et al., 2019).

Voltage sensing by MEA

Mechanism of voltage recording for cardiotoxicity screening

Like Neurons, cardiomyocytes are also electrogenic cells capable of generating action potentials (APs). An AP represents a rapid change in the electrical voltage between the intracellular and extracellular space, separated by the cell membrane, and is thus referred to as the transmembrane potential. The gold standard for direct measurement of the transmembrane potential is still Patch clamp (Syren et al., 2025; Yip et al., 2024). Although automatic whole-cell patch clamp setups are available (Suk et al., 2019), simultaneous recording of multiple intracellular action potentials is tedious and low-throughput. MEAs are a potential alternative that can achieve high throughput, field potential measurements, and direct AP measurements using novel technologies such as penetrating MEA setups or the open-cell technique. Microelectrode arrays (MEAs) offer a high-throughput alternative for electrophysiological monitoring. They enable field potential tracking and direct-action potential (AP) measurements via penetrating configurations or open-cell techniques (Obien et al., 2015). However, intracellular recording requires three-dimensional, sub-micron electrodes to penetrate the cell membrane and access the cytoplasm without inducing significant cytotoxicity; target biological cells generally measure only tens of micrometres in diameter. Typically, achieving this intracellular interface necessitates secondary procedures such as mechanical pressure, negative suction or localised electroporation to introduce the microscale or nanoscale sensor into the cell interior (Jones et al., 2020). Despite initial success with these techniques, these platforms remain limited to research prototypes. Significant challenges regarding reproducibility of recordings have prevented 3D MEAs from becoming the gold standard, leaving the market without any commercially available systems. Consequently, as intracellular action potential tracking remains experimentally complex and demanding, researchers routinely rely on alternative extracellular MEA configurations to indirectly capture action potentials via external field potentials on the outside of excitable membranes. The simplest voltage measuring mode of a MEA to record the electrical activity of cardiomyocytes is field potential (FP) recording. The intracellular electrodes are omitted, and the electrical signal is measured exclusively from extracellular electrodes. Consequently, it is crucial to understand the link between the AP and the extracellular FP, which is given in the following passages.

All ion movements influence the intra- and extracellular electrical potential, since these also represent an electrical current, similarly to electron movements in metals. Therefore, the ion currents through the membrane (transmembrane currents) during depolarization or repolarization of the cardiomyocyte lead to changes in the intra- and extracellular electrical potential (Buzsáki et al., 2012).

The transmembrane current I_{mem} cardiomyocytes can be electrically modelled using the Hodgkin–Huxley model (Forouzandehmehr et al., 2021). The corresponding equivalent circuit model is shown in Figure 5A and was initially introduced by Hodgkin and Huxley (H-H) (Hodgkin and Huxley, 1952) to model the properties of neuronal membranes.

The membrane of human biological cells consists of a lipid bilayer, which is, by itself, impermeable to most molecules and ions. From an electrical point of view, this acts as a capacitor C_m. To enable transfer of substances across the membrane, different active and passive protein channels, leakage protein channels, and protein pumps are integrated into the bilipid membrane. When a cell fires an action potential, gated protein channels open and close. In the H-H model, protein channels are represented by their individual conductance g_i (gated channels - non-linear, time and voltage-dependent, with i being the index of the transported molecule) and g_L (linear - leakage channels). The electrochemical gradients driving the ion flow are represented as voltage sources V_i and V_L. Finally, the ion pumps are represented as a current source I_P.

Using this model, the total current through a membrane patch consists of the ion currents through all gated and leakage channels, as well as the capacitive current through this patch, as shown in Equation 1.

$$I_{mem} = C_m \frac{\partial V_{mem}}{\partial t} + g_i^*(V_m - V_i) + g_L^*(V_m - V_L) + I_P \qquad (1)$$

Using the assumption of a simplified fiber geometry, the one-dimensional cable equation from cable theory can be used, with the transmembrane current corresponding to the radial current proportion of the cable. Therefore, I_{mem} can be denoted as proportional to the second derivative of V_{mem} in space and the proportional factor being the diffusion parameter D (Alonso et al., 2025). For simplification, the AP can be assumed as a fixed potential traveling wave with a constant speed v, the second derivative in

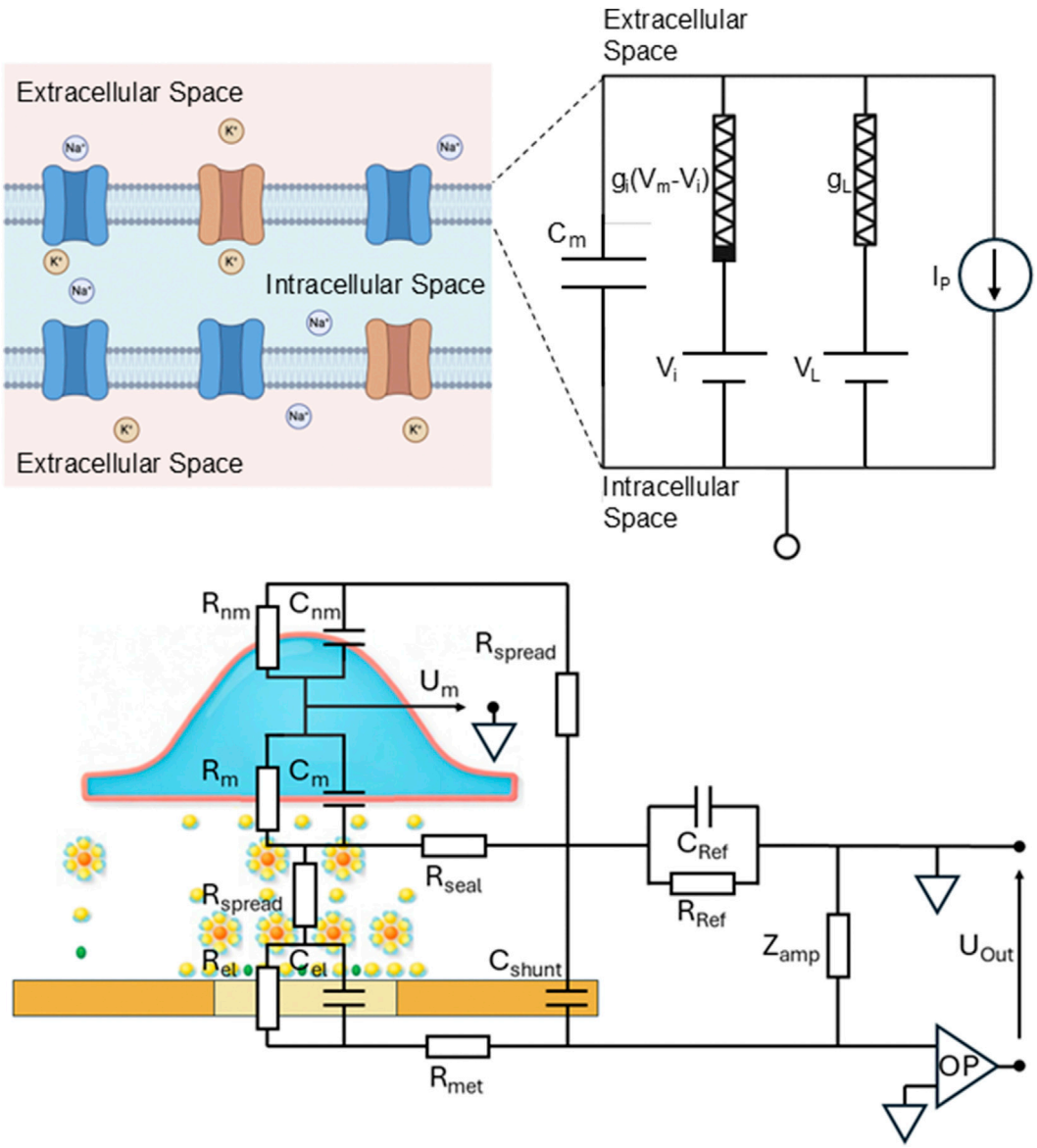


FIGURE 5
Schematic and equivalent circuit diagram of cell membrane and cell-electrode interface. **(A)** Schematic of cell membrane and its equivalent circuit diagram. **(B)** Schematic cel electrode interface including Helmholtz layer and its equivalent circuit diagram. Previously published material has been adapted: Equivalent circuit of the Hodgkin–Huxley model — adapted from doi: 10.3390/math11030714, licensed under CC BY 4.0. Equivalent circuit of the electrode–cell interface — adapted from Shaik et al., doi: 10.1007/978-981-10-8953-4_22 (Springer); with permission from the rights holder.

space can be replaced by the second derivative in time (McColgan et al., 2017) as shown in Equation 2.

$$I_{mem} = D\frac{\partial^2 V_{mem}}{\partial x^2} \rightarrow I_{mem} = \frac{1}{v^2} D\frac{\partial^2 V_{mem}}{\partial t^2} \quad (2)$$

Combining equations (Equation 1) and (Equation 2) I_{mem} results in (Equation 3).

$$I_{mem} = D\frac{\partial^2 V_{mem}}{\partial t^2} = C_m\frac{\partial V_{mem}}{\partial t} + g_i^*(V_m - V_i) + g_L^*(V_m - V_L) + I_P \quad (3)$$

When it comes to the recorded voltage at the MEA electrode, the distance between the transmembrane currents and the electrode becomes a key factor. This must be considered for cardiotoxicity screening, to choose a MEA conception that provides a signal capable of recording the parameters of interest. The recorded signal is determined by the following situations, which are reviewed in detail in the following passages: (1) The recorded signal is proportional to the second derivative of the membrane voltage $\frac{d^2V_m}{dt^2}$ if the cell is not or only sparsely attached to the electrode. The recorded signal is proportional to the first temporal derivative of the membrane voltage $\frac{dV_m}{dt}$ if the cell is attached with loose contact (loose-cell). (2) The recorded signal is directly proportional to the membrane potential in the case of very strong contact and low junctional resistance. The derivative of a signal indicates the incline and decline of the signal, resulting in a

dampening effect and potential loss of information. Some of the possible limitations of extracellular field potential recordings are discussed in (Spira and Hai, 2013).

Cells that are not directly attached or only very loosely attached to the electrodes are surrounded by a liquid, ion-containing medium. Therefore, the volume conduction theory describes the electric potential at the electrodes, e.g., assuming a point current source (Meyer et al., 2020). According to volume conduction, the recorded voltage will be proportional to the transmembrane current. Therefore, the recorded field potential will be proportional to the second derivative of the AP, which results in a characteristic triphasic shape. For large, distant electrodes, the electric potential decreases by the square of the distance, resulting in small signals. Additionally, the potential changes of all transmembrane ion currents of all cells will contribute to the potential at the distant electrode. Consequently, the recorded signal will be a superposition of these individual signals at the microelectrode. This is especially important when 3D heart organoids are measured and is also the fundamental principle for the ECG signal. More complex calculations can include inhomogeneous media (Bédard et al., 2004) and the use of a current dipole source instead of a monopole source (Mechler and Victor, 2012).

In the case of single cardiomyocytes directly attaching to the electrode using their cell adhesion molecules (CAMs) (such as cadherins, integrins), the point-contact model can be used, which was originally introduced by Fromherz et al. (1991). A simplified version by Shaik et al. (2022) is shown in Figure 5B. The intracellular membrane potential is referred to as U_m and the measured external potential as U_{out}. The measurement is carried out between a reference electrode, represented as a parallel circuit of a resistor R_{Ref} and a capacitor C_{Ref}. The small membrane patch that is in contact with the electrode is referred to as coupling or junctional membrane (Spira and Hai, 2013); the rest, not in contact with the electrode, is referred to as non-coupling or non-junctional membrane. Both membrane parts are electrically represented by a parallel circuit of a resistor (R_m *or* R_{nm}) to model the ion currents and a capacitance (C_m *or* C_{nm}) to model the capacitive current. Underneath the junctional membrane, the membrane-electrode cleft is filled with culture media, which electrically behaves as an aqueous electrolyte, represented as R_{spread}. Because of the close contact between the cell membrane and the electrode, the ion flow between the coupling- and non-coupling is restricted, which is represented as Rseal. The electrode with the Helmholtz layer is modeled using the Randles circuit, which is a parallel circuit of the Resistor R_{el} and the Capacitor C_{el}. The resistance of the electrode and interconnect metal is represented as R_{met} and the capacitance of the insulating layer as C_{Shunt}. The resistance of the measurement amplifier R_{OP} is included in Z_{amp}. By neglecting R_{nm}// C_{nm} , the different components can be combined into a simplified equivalent circuit shown in Figure 5B. The membrane impedance Z_m, the electrode impedance Z_{el} , and the amplifier impedance Z_{amp} can be calculated by Equations 4–6 respectively, with the angular frequency ω. Finally, the connection between the measured voltage U_{Out} and the internal membrane potential U_m is given by Formula 7.

$$Z_m(\omega) = \frac{R_m}{1 + i^*\omega^*C_m^*R_m} \tag{4}$$

$$Z_{el}(\omega) = \frac{R_{el}}{1 + i^*\omega^*C_{el}^*R_{el}} + R_{Spread} + R_{met} \tag{5}$$

$$Z_{amp}(\omega) = \frac{R_{OP}}{1 + i^*\omega^*C_{Shunt}^*R_{OP}} \tag{6}$$

$$V_{Out} = V_m^* \frac{R_{seal}^* Z_{amp}}{(R_{seal} + Z_m)^*\left(Z_{amp} + Z_{el}\right) + R_{seal}^* Z_m} \tag{7}$$

Equivalent circuit of the electrode-cell interface (Shaik et al., 2022).

The tightness of the electrode-cell contact is a key parameter as it determines if the recorded signal is proportional to the AP (V_m) or the first derivative of the AP ($\frac{dV_m}{dt}$). Fromherz et al. (1991) could show that the recorded signal directly corresponded to the AP (V_m) if a tight seal, corresponding to a high R_{seal}, is present. In contrast, loose attachment leads to recording of the first derivative of the AP ($\frac{dV_m}{dt}$). Furthermore, Shmoel et al. (2016) showed that the relation between R_m and C_m is also significantly influenced by whether the recorded signal is proportional to V_m (referred to as in-cell recording) or $\frac{dV_m}{dt}$ (referred to as juxta cellular recording). A high R_m leads to juxtacellular ($\frac{dV_m}{dt}$) recording, while a low R_j leads to in-cell (V_m) recording. In their work, they demonstrated the use of mushroom-shaped 3D electrodes to achieve tight contact and to decrease R_m. Additionally, their work proposed that further enlargement of the mushroom head and the effective mushroom head surface using coatings, may further decrease R_m to enable in-cell-like recordings.

A similar principle of the reduction of R_m, while remaining a high R_{seal} is also used in MEAs with 3D electrodes that are designed to penetrate the cell. An Additional possibility to directly measure the AP, based reduction of R_m, is temporary and local permeabilization of the membrane using electro- (Xu et al., 2025) or optoporation (Iachetta et al., 2023). To facilitate a direct, parameter-driven comparison between these recording modalities and highlight their respective engineering trade-offs, a comprehensive breakdown of their hardware requirements, excitation configurations, and readout capabilities is consolidated in Table 4.

Field potential features for cardiotoxicity screening

MEA offers sub-millisecond temporal resolution for recording FPs that showcase various parameters for analysis, including beat rate, field potential duration (FPD), and spike amplitude. Each of these parameters are akin to heart rate, QT interval, and QRS amplitude in an ECG signal (Gintant et al., 2019). The FP gives the following endpoint information about the action potential: 1) repolarization, 2) depolarization 3) beating time 4) arrhythmia occurrence and 5) propagation. As suggested, several studies have demonstrated the correlation between field action potential (fAP) data obtained through extracellular measurements using micro-electrode arrays (MEAs) and both action potential and ECG. In a study published in 2004, researchers demonstrated that the critical factor contributing to QT prolongation is mirrored by the extension of field action potentials (fAP) and can be detected in micro-electrode array (MEA) recordings from cardiac myocytes (Meyer et al., 2004).

The transmembrane potential across the cardiomyocyte's lipid bilayer can cause an action potential when it reaches a threshold due to stimuli or other inputs. We refer to this signal coming from a putative action potential a "Spike" which in extracellular recording are distinguished as voltage signals that

TABLE 4 Quantitative comparative analysis of hardware specifications, electrical excitation boundaries, and readout performance metrics across extracellular, intracellular (electroporation-mediated), and patch-clamp electrophysiological detection modalities.

Parameter Category	Parameter	Extracellular detection (passive)	Intracellular detection (electroporation)	Whole cell patch clamp (gold standard)
Hardware parameters	Electrode Material	• Charge Injection Capacity (CIC) is mostly irrelevant. • Standard metals like Au (CIC 0.2 mC/cm^2) and Pt (CIC 0.83 mC/cm^2) are sufficient. Optional: Ti_3N_4 or porous PEDOT:PSS can be used to reduce the electrode impedance (Liang et al., 2021; Zheng et al., 2021)	• High-CIC materials required to avoid electrochemistry (e.g., hydrolysis) at the electrode • Low impedance coatings required for a high cell coupling coefficient (Spira and Hai, 2013) • Advanced electrode materials required: PEDOT:PSS (CIC 1.9–2.71 mC/cm^2) (Ganji et al., 2017), wet spun Carbon Nano Tube (CNT) fibers (CIC 6.52 mC/cm^2). An overview of electrode materials is shown in (Zheng et al., 2021)	• Charge Injection Capacity (CIC) is mostly irrelevant. • Typically, Ag/AgCl electrodes inside an electrolyte-filled glass pipette (Inayat et al., 2013)
	Cell-Electrode Coupling	• High seal resistance (cleft isolation) is beneficial to prevent signal shunting, which lowers the SNR Typical Rseal range: 10^6 Ω (Spira and Hai, 2013; Hai et al., 2009)	• Extremely good seal required via advanced topographies (e.g., 3D vertical graphene) to minimize attenuation and ion leakage (Liu X. et al., 2023) • Typical RSeal range: 10^8 Ω	• Gigaseal required (highest demand on the seal) (Inayat et al., 2013) • Typical RSeal range: 10^9 Ω range (Spira and Hai, 2013)
Electrical Excitation parameters (for electroporation)	Signal Amplitude	• Passive recording (0 V/0 A).	• Voltage-clamp: typical 0.5 V–4.0 V (Qu et al., 2025) • Current-clamp: typical 0.5–3.0 nA • indicates that increasing field strengths increases the pore density. 0/0/00 0:00:00 AM	• Passive (0 V/0 A); optional high-voltage zapping to open membrane (Inayat et al., 2013)
	Excitation Signal Shape	• Not active pulsing required.	• Typically, biphasic pulse trains, e.g., 20 biphasic pulses with a width of 200 µs in 1s (Xie et al., 2012) • indicates that an in increase in the pulse width enlarges the pore diameter. • Optimizing the pulse width together with the amplitude is critical to prevent cell death (Qu et al., 2025)	• No active pulsing required (except for zapping).
Readout parameters	Readout signal shape	• Proportional to the 1st (high seal, typical for MEAs or 2nd (loose seal) time derivative of the intracellular AP (see previous passages). (Guo, 2020)	• Directly proportional to the absolute transmembrane potential but attenuated. DC signal such as the resting potential are attenuated the most, due to the AC coupled measurement.	• Measurement of the full AP morphology.
	Signal amplitude	• µV range 10–500 µV; • Heavily attenuated compared to the AP • Dependent on electrode size, material, and coupling (Liu X. et al., 2023)	• mV range (+6 [10.3389/fneng.2012.00021] to +11.8 mV (Fendyur and Spira, 2012; Xie et al., 2012) • attenuated compared to the AP, described by the Cell Coupling (CC) factor • Key factors for a high CC are: High seal resistance, low electrode impedance, a low junctional membrane resistance a low stray capacitance [10.1038/NNANO.2012.265] and the number of electroporation cycles (Liu X. et al., 2023)	• Measurement of the full AP amplitude (−90 mV to +30 mV (Szedlak et al., 2023)

(*Continued*)

TABLE 4 Continued

Parameter Category	Parameter	Extracellular detection (passive)	Intracellular detection (electroporation)	Whole cell patch clamp (gold standard)
	Extractable features	• Extracellular FP parameters such as: spike firing rate/beat period, spike amplitude, field potential duration (FPD) (Kito et al., 2024) and Inter-spike interval.	• Sub (AP)threshold signals (Spira and Hai, 2013) • AP parameters such as APD_{50} and APD_{90}, Depolarization time and the Action Potential Amplitude (APA) (Lee et al., 2024) • Relative resting potential voltage changes	• All morphological parameters and absolute voltage levels determined directly.
	Readout times	• Continuous; limited only by amplifier sampling rate (10–20 kHz).	• Transient up to 83 min; dictated strictly by biological pore repair kinetics (Liu X. et al., 2023)	• Transient ~35 min (standard) up to 12 h using Perforated patch recording (Husch et al., 2011)
	Throughput	• High throughput for 2D cell layers and 3D constructs.	• High throughput mainly for 2D cell layers.	• Low throughput; restricted to single/few cells.
Summary	Pros and Cons	Pro: continuous recording and high throughput for 2D/3D networks. Con: heavily attenuated time-derivative signals that limit the number of extractable parameters	Pro: high throughput (2D) yielding proportional transmembrane potential and AP kinetics. Con: requires complex electrodes, strict pulsing limits, and readouts are transient.	Pro: provides full, unattenuated AP morphology and exact voltages. Con: low throughput, transient, and demands strict Giga seal.

exceed a threshold (Feher, 2012). In intracellular action potential recordings, first we observe a positive signal that is followed by a negative volley in neurons and in cardiac cells there is a positive signal followed by a brief repolarization and a longer plateau. In extracellular field potential (EAP) recording, there is a large positive spike (corresponding to the initial rapid sodium influx (Figure 6) which is followed by a negative spike associated to the potassium efflux from the cardiomyocytes (Dipalo et al., 2017).

FPs are distinguishable only when the electrodes are located at the vicinity (about 100 um) of the spike origin, usually around the cell; intracellular action potentials are on the other hand detected when electrodes access the inside of the cells (Meyer et al., 2020). The intracellular action potential duration corresponds to the duration of QT signal on an ECG signal. In experimental and computational work, it has been shown that three signal shapes can be recorded from cells based on their cell adhesion and geometry of the electrode. An AP-like shape, a second derivative shape and an intermediate waveform depicted in (Figure 6). The second derivative signal shapes are recorded from cells or tissue slices near the electrode and not from adherent cells in close contact to the electrodes. In the case of the AP-like shape, at greater than 10 mV in amplitude, the microelectrodes can capture over 10% of the full intracellular voltage change in the AP-like shape. The most observed signal shape is the second derivative shape which can change in the presence of EADs, and ectopic beats as depicted in (Figure 6). Contraction of cardiomyocytes cannot be directly derived from a FP signal. However, microelectrode arrays can be used for directly investigating the contractility of the cardiomyocytes by measuring impedance (Figure 6).

Mechanism for impedance-based cardiotoxicity measurement

While field potentials are mainly recorded from electrically excitable cells that can generate action potentials, impedance measurements are used regardless of the electrical activity of the cell. In EIS measurements of cardiomyocytes, two main mechanisms are of interest: (1) Determination of the contractility of the cardiomyocytes and (2) direct determination of the viability using the EIS. The underlying mechanisms are explained in this chapter.

The Impedance provides data on the conductive and dielectric properties of the materials between a pair of electrodes. These materials are modelled as an electrical two-terminal circuit element, where the impedance links the voltage between the terminals and the current through the element. Simplified, the impedance is the total opposition that limits the flow of charge carriers (current), which are driven by a sinusoidal voltage. To avoid the use of differential equations, impedance relies on complex AC circuit analysis. In general, e. g., impedance measurement of cells, the complex impedance $\underline{Z}(\omega)$ is frequency-dependent due to polarization effects. Consequently, the complex impedance $\underline{Z}(\omega)$ is given by the complex $\underline{v}(\omega, t)$ and the complex current $\underline{i}(\omega, t)$ as shown in Equation 8, with their maxima $\hat{v}(\omega)$ and $\hat{I}(\omega)$, the effective values V (ω) and I (ω) and their phase angles $\varphi_v(\omega)$ and $\varphi_i(\omega)$.

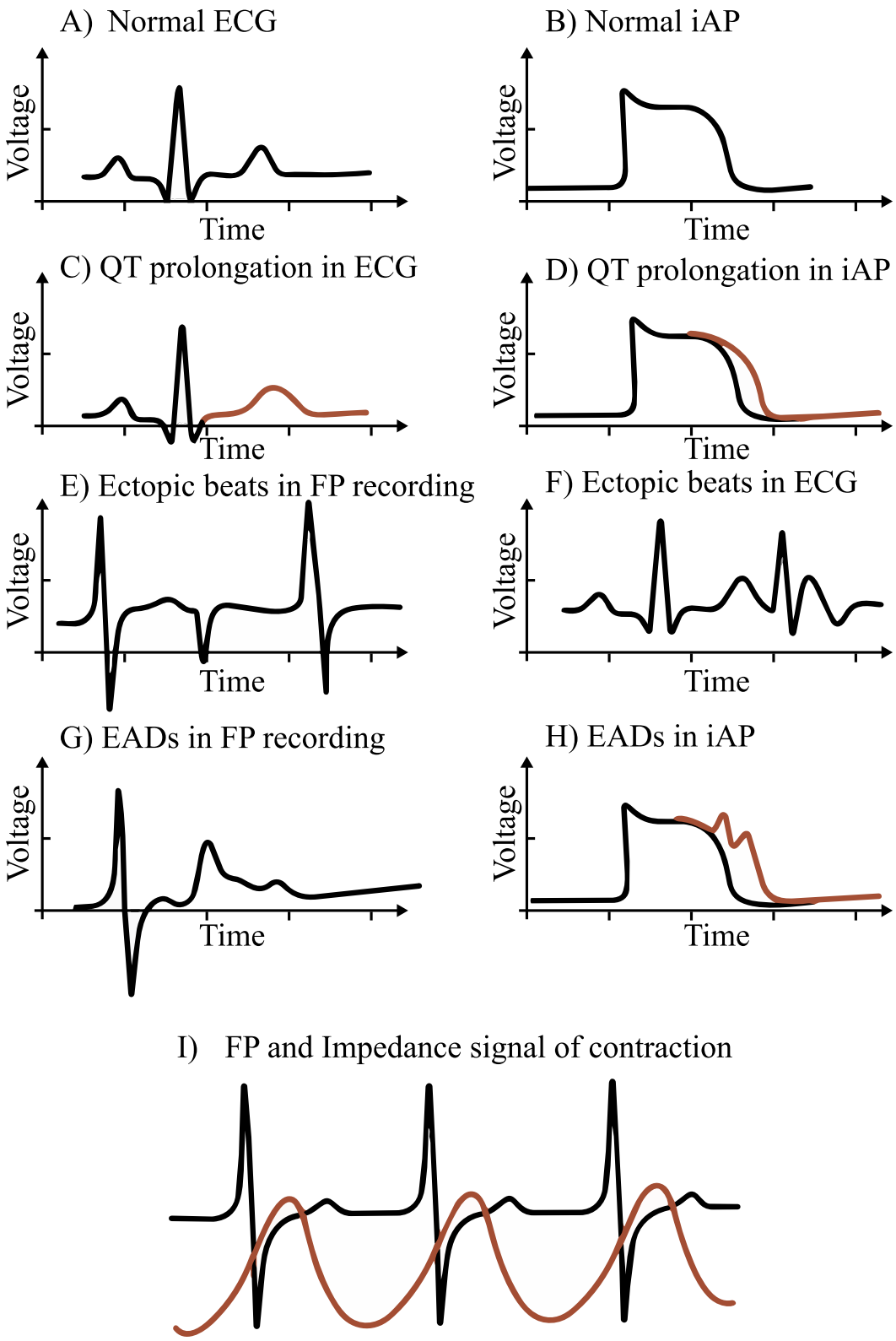


FIGURE 6
Proarrhythmic markers in different cardiac recording modalities. Comparison of normal and pathological cardiac signals: **(A,B)** Normal ECG and intracellular action potential (iAP). **(C,D)** QT/APD prolongation (red overlay indicates extension). **(E,F)** Ectopic beats in field potential (FP) and ECG recordings. **(G,H)** Early afterdepolarizations (EADs) in FP and iAP (red trace shows characteristic secondary depolarizations). **(I)** Simultaneous FP (black) and impedance-based contractility (orange) recordings demonstrating electromechanical coupling. These parameters serve as key cardiotoxicity biomarkers in MEA-based screening of cancer therapeutics.

$$\underline{Z}(\omega) = \frac{\underline{v}(\omega,t)}{\underline{i}(\omega,t)} = \frac{\hat{v}(\omega)}{\hat{I}(\omega)} * e^{i(\varphi_v(\omega)-\varphi_i(\omega))} = \frac{V(\omega)}{I(\omega)} * e^{i(\varphi_v(\omega)-\varphi_i(\omega))}$$
$$= |Z(\omega)| * e^{i\varphi(\omega)} \quad (8)$$

With the angular frequency $\omega = 2*\pi*f$ and the phase shift $\varphi(\omega)$ between $\underline{v}(\omega,t)$ and $\underline{i}(\omega,t)$.

The frequency dependence of $\underline{Z}(\omega)$ is the basis for electrochemical impedance spectroscopy (EIS), where $\underline{Z}(\omega)$ is measured at a spectrum of frequencies. Typical representations of $\underline{Z}(\omega)$ are the Nyquist and the Bode plot. The Nyquist plot shows the imaginary part Im ($\underline{Z}(\omega)$) over the real part of Re ($\underline{Z}(\omega)$). The Bode plot consists of two plots, one showing the impedance magnitude $|Z(\omega)|$ over the frequency and one showing the phase $\varphi(\omega)$ over the frequency f.

In biological samples, such as cardiomyocytes, the frequency dependence of the EIS spectrum is linked to cell-electrode adhesion, cell morphology, and viability. From low to high frequencies, the cellular dispersion regions are referred to as alpha (10 Hz–10 kHz), beta (10 kHz–100 MHz), delta (100 MHz–10 GHz), and gamma (>10 GHz) (Abasi et al., 2022). To understand the cardiomyocyte-specific impedance changes, such as caused by beating, it is crucial to understand the alpha and beta dispersion mechanisms.

The alpha and beta dispersion mainly arise from the impermeability of the biological cell membrane for most molecules, which acts as an electrical insulator. This leads to a capacitive behaviour of the membrane. Furthermore, the outer surface of the cell membrane carries negative charges. It is therefore surrounded by an adsorbed and a diffuse layer of positively charged counterions, known as electric double layer (EDL). The three main mechanisms for EDL formation at the cell membrane are adsorption, dissociation, and ion pumps.

When an external AC field is applied, all ions (media and EDL) will oscillate back and forth, driven by the electric force. The travel distance of the ions is highly dependent on the frequency due to their limited mobility. If the ions encounter an impermeable interface on their way, e.g., the cell membrane, they will accumulate or move tangentially along the surface. This leads to ion- and interfacial polarization, a separation of the centres of the positive and negative charges, creating a dipole.

In the low-frequency alpha dispersion range, the current will be effectively blocked by the cellular membrane, as the membrane capacitance charges quickly in the slowly changing external field. Consequently, most of the current will flow in the interstitial space around the cell. The exact reason for the low-frequency alpha dispersion is not fully known (Grimnes and Martinsen, 2010; Chakraborty et al., 2020). One of the mechanisms considered to have a major effect is polarization caused by the displacement of the EDL ions, initially introduced by Schwan (Schwan, 1957). The EDL ions around the cell need a very high external field to escape from the surface, whereas tangential movement along the membrane can be achieved with much lower fields. When a sufficiently low-frequency external field is applied, the EDL ions have sufficient time to move along the membrane, leading to polarization (Grimnes and Martinsen, 2010). Consequently, the polarization is at maximum when an equilibrium between the forces, mainly diffusion and external electric force, is reached on the EDL ions. With increasing frequency, the ions will not have enough time to achieve maximum polarization, resulting in alpha dispersion. In practical measurements, the ions will also flow in the direction of the electrode and accumulate there. Consequently, these electrode polarization effects are often largely dominating in the alpha dispersion frequency region and need to be corrected (Kalvøy et al., 2011).

With increasing frequencies, the travel distance will be insufficient for polarization by tangential movement along the membrane. Consequently, the Maxwell-Wagner interfacial polarization at the membrane is the dominant effect in the beta dispersion frequency region (Martinsen et al., 2002). Aligned with the external field, ions will undergo oscillations normal to the cell membrane, resulting in charge accumulation at the membrane and polarization. The counterions on the inside are moving accordingly, thus, electrically speaking, a displacement current flows through the membrane. This leads to a decreased impedance and underlies the capacitive behaviour of the cell membrane. With increasing frequency, the ions cannot fully charge the membrane capacitor, and the polarization decreases, resulting in beta dispersion.

Every contraction of the cardiomyocyte leads to a change in the cell morphology and the distance in the cell-electrode gap. As shown by Giaever and Keese (Giaever and Keese, 1991), the EIS spectrum is highly sensitive to changes in the cell-electrode distance, capable of measuring distance changes of 1 nm. Another study by Sarró et al. (2016) shows the effects of the cell morphology on the EIS spectrum. Depending on the cell morphology, the ions must travel different distances for polarization, resulting in altered relaxation times. The relaxation time is directly linked to the minimum in the phase Bode plot. Consequently, large cells exhibit a shift of the phase minimum to lower frequencies compared to smaller cells.

Besides EIS measurements for contractility measurements, it can also be used to assess the viability of the cells. Depending on the cell death mechanism (Galluzzi et al., 2018), dying cells change the surface charge (Galluzzi et al., 2018), morphology, e.g., shrinking during early apoptosis (Elmore, 2007) or swelling during oncosis (Van Cruchten and Van den Broeck, 2002) and membrane capacitance Consequently, the viability can be evaluated using EIS, with dying cells typically showing a decreased impedance magnitude, e.g., due to membrane disruption during necrosis (Brantlov et al., 2025). EIS measurements also show high potential for measuring 3D cell cultures, such as spheroids and organoids (Chawla et al., 2018).

Compared to other cell types, cardiomyocytes exhibit rapid changes in their electrical behaviour due to their beating activity. This places particular demands on the EIS measurement, as complex AC circuit analysis requires the use of sinusoidal excitation waveforms and the system to be linear and time invariant (LTI) and causal. Linear means that the current and voltage are only linearly connected to the impedance, e.g., a double voltage leads to a double current. Time-invariant describes the application of a specific excitation, which will always lead to the same impedance, independent of time. Lastly, causal systems are only affected by the excitation, not any future events (Hallemans et al., 2023). Cardiomyocytes are clearly non-LTI systems, due to nonlinear membrane protein channels and time-variant physiological processes such as growth and contraction during beating. To overcome the non-linearity of biological samples, low amplitude voltages are applied for excitation, keeping the system in a pseudo

linear working range (Stupin et al., 2021) but limiting the signal to noise ratio (SNR). To achieve a pseudo time invariance during the recording, the measurement of the complete spectra must be faster than the cell dynamics to observe a constant state. To speed up the measurement, multi-sine approaches and broadband excitation signals, such as square wave approaches, are available (Sanchez et al., 2013). Broadband signals can be transformed into sinusoidal waves of different frequencies that sum up to the original signal using the Fourier transformation. The impedance at specific frequencies can then be calculated using the individual sinusoidal frequency components. To test if the recorded impedance spectrum fulfils all requirements, the Kramers-Kroenig equation can be applied to the data (Hallemans et al., 2023).

EIS measurements inherently record a superposition of different effects, such as changes in cellular, media, and electrode surface properties. To distinguish these effects from the single EIS recording, typically equivalent circuit models (ECMs) are used. First, an equivalent circuit describing the measured situation is chosen, Typical ECMs for biological tissue are the Cole-Cole and Fricke-Morse model, which are described in detail in (Abasi et al., 2022) and (Kasiviswanathan et al., 2020). The individual components of the ECM are then altered, e.g., using a numerical solver, until the resulting EIS spectra of the model fits the real measurement. Consequently, the various ECM components can be examined individually.

MEA: from cardiotoxicity to cancer induced cardiotoxicity assessment

MEAs have applications in evaluating cardiotoxicity, one of the pioneering works on MEA and cardiotoxicity was done by using embryonic chicken ventricular cells to study reference substances (E4031, amiodarone, quinidine and sotalol) with reported QT interval prolongation. It was interesting that verapamil, a typical "false positive" when using the hERG assay does not cause any field potential prolongation using the MEA assay. The hERG assay is a screening tool for drug-induced QT prolongation risk which measures the blockade of the hERG potassium channel. It is mostly done in heterologous expression systems along with patch clamp or automated patch clamp (Garrido et al., 2020). Radioligand binding assays and fluorescence based assays also exist for hERG blockade screening along with *in silico* hERG modeling (Garrido et al., 2020). This is in fact due to the reason that MEA assays reflect the full range of mechanisms involved in cardiac action potential regulation unlike repolarization of one channel in hERG assays. Due to the transient culture of adult cardiomyocytes, scientists have turned to human induced pluripotent stem cell derived cardiomyocytes and more recently organoids (Hsueh et al., 2023; Matsa et al., 2011). Undifferentiated hiPSCs can, in theory, be cultured indefinitely, the supply of human CMs is likewise unlimited. Significantly, patient-specific hiPSC-CMs can be produced to study drug effects in the context of a particular risk factor or disease mutation.

One of the early incidences of the use of hiPSC-derived cardiomyocytes and MEAs was published in Navarrete et al. (2013) using a comprehensive panel of drugs from various classes. Navarrete et al. (2013) study revealed that hiPSCs derived cardiomyocytes express nearly all known ion channels and subunits that align with the characteristics of adult human ventricular tissues. This includes action potentials (APs) that resemble the electrophysiological properties of ventricular myocytes, even though hiPSC-CMs exhibit immature morphology and spontaneous beating. By forming embryoid bodies (EBs), a multicellular preparation, they circumvented the limitations associated with reductionist single-cell preparations and intrinsic cell differences (Navarrete et al., 2013).

Using this system, they successfully demonstrated the reliable identification of drugs prone to causing false negative and false positive Torsades de Pointes (TdP), a ventricular arrhythmia, and showcased the sensitivity of the assay to ion channel blockers. Moreover, they highlighted the effectiveness of the MEA/hiPSC-CMs assay in both detecting and quantifying arrhythmic events like Early Afterdepolarizations (EADs) and ectopic beats, which are the actual underlying mechanisms behind TdP (Navarrete et al., 2013) and are all evident on MEAs alone.

Interestingly, to avoid the reductionistic approach of single mono layer of hiPSC-CMs, several research groups have recently worked on developing 3D constructs of these cells to promote tissue maturation, higher complexity (Schmidt et al., 2023; Hofbauer et al., 2021) and have included endothelial vascularization, an epicardial layer (Meier et al., 2023), and an internal cavity. Recently, Schmidt et al., developed cardioids resembling different regions of the heart (atrial, ventricular, atrioventricular and outflow tract) and showed that each cardioid subtypes develop distinct ion channel expressions by day 9.5. They put whole cardioids on CMOS-MEAs (3 Brain, Zurich, Switzerland) which have 64×64 electrode grid and measured the field potential at high spatial resolution. They observed field potential diversity across cardioid subtypes. For left ventricular cardioids, they showed a more homogenous signal propagation than atrial and atrioventricular cardioids. They also showed that the left ventricular and right ventricular cardiomyocytes have longer action potential durations than atrial when conducting voltage imaging on 2D cultures. The cardioids were also analyzed using patch clamp technique and showed that the upstroke velocity and amplitude of the action potentials resembled the most advanced *in vitro* models and specific CMs were electro physiologically homogenous. Their conclusion was that the electrochemical signaling of cardioids is diverse and fetal-like (Schmidt et al., 2023). Therefore, MEAs have been interfaced with both 2D monolayer of biological models of cardiomyocytes (HL-1 cell line, primary neonatal rat cardiomyocytes and hSC-CMs) and 3D heart organoids.

Beyond the extracellular field potential recording, MEAs can also access intracellular- like recording of action potentials. This is done by transiently permeabilization of the patch of membrane facing the electrode. This is done by electroporation, optoporation, or spontaneous fusion with a nanostructure; this gives the electrode partial access to the intracellular compartment and records an attenuated intracellular-like action potential. This enables the possibility of potentially recording true AP morphology and duration. Theoretically in most mentioned methods for permeabilizing the membrane, the membrane reseals afterwards, making the approach transient and repeatable.

Intracellular recording of action potential has been reported previously, and some instances are by using nanopillars and nanoscale electroporation (Xie et al., 2012), electroporation with

planar MEA (Xu et al., 2022), nano branched microelectrode arrays 3D plasmonic nanoelectrodes (Dipalo et al., 2017). In these instances the intracellular-like signals are recorded from single cells at a time. To break this constraint, in a 2020 paper, a CMOS nanoelectrode array was used to achieve network level neuronal activity recording (iMEA or intracellular MEA). In most cases-controlled electroporation with and without nanopillars enables intracellular-like recordings. The different approaches differ mainly in how the operation is achieved and at what scale. Comprehensive integration of this intracellular capability with contractile and morphological read-out remains an emerging direction beyond the primary scope of the present review.

It has also become possible to use MEA in 2-point, 3-point and 4-point measurement setups for impedance measurement of contractility and physical status of cardiomyocytes. Hu et al. (2015) developed a pair of interdigitated electrodes complemented with a pair of planar microelectrode arrays and reference electrodes for developing a bifunctional MEA. Using this MEA and HL-1 atrial cell lines they successfully showed the contractility measurement and field potential recording of cardiomyocytes with one chip (Gu et al., 2019). In a lab on a chip publication, another configuration of interdigitated electrodes and planar circular microelectrode arrays were used for this purpose with hSC-CMs (Qian et al., 2017). Please refer to this review (Wei et al., 2020) for reviewing different methods for monitoring contractility of cardiomyocytes.

MEAs have also been used for studying chemotherapeutic effects such as doxorubicin and immunotherapies such as Tyrosine kinase inhibitors. In one study, MEA and hiPSC derived cardiomyocytes were used to study the effect of Doxorubicin. Xu et al. (2022) used ExCell which is a dual functional extracellular chip with microelectrode arrays and interdigitated electrodes for evaluation of doxorubicin toxicity on cardiac cells. This multi-well chip contains four chambers, and it integrates various parameters such as amplitude, beating rate, beat-to-beat variation, Ca2+ flux derived from extracellular field potential, and cell substrate impedance. These combined parameters provide a comprehensive profile of doxorubicin-induced cardiotoxicity, helping to identify cardiac failure, arrhythmias, and even cellular death, aligning with previously published research findings (Iachetta et al., 2023; Xu et al., 2022).

In another paper (Iachetta et al., 2023), long term *in vitro* recording of cardiac action potential in response to doxorubicin was reported. They combined commercial, transparent MEAs from multichannel systems with optoporation to obtain intracellular-like action potential signals from iPSC-derived cardiac cells. Cardiotoxicity caused by doxorubicin is due to chronic exposure of patients to this chemotherapeutic. Therefore, in their innovative approach, the researchers recorded these signals for up to 24 days after drug treatment. They observed a gradual reduction in action potential duration (APD) starting at 21 days post-drug treatment for the lower tested concentration and at 19 days post-treatment for the higher concentration. Additionally, they noted a decrease in spike amplitude for both tested concentrations. These findings were consistent with the results reported by Maillet et al., which included an increase in beat rate, a decrease in spike amplitude, and a shortening of the corrected field potential duration (cFPD) observed after 20 h of treatment.

In a recent publication in nature biomedical engineering, Ghosheh et al., 2023, used multi-chambered vascularized human cardiac organoids (Qin et al., 2024) and custom-built optical permissive MEAs. They showed that the chemotherapeutic mitoxantrone induces arrhythmia through disruption of electro-mitochondrial coupling in cardiomyocytes which is driven by mitochondrial calcium oscillations. And further showed that this process can be reversed partially by co-administration of metformin (Qin et al., 2024).

Other than optical methods for monitoring calcium oscillations and electromechanical coupling, several papers explored impedance measurement of contractility and field potential via MEA in response to chemotherapeutics and immunotherapies (Jiang et al., 2023; Wei et al., 2019). Using ExCell MEA, Ping Wang evaluated the toxicity induced by doxorubicin on cardiomyocytes (Wang et al., 2020). They showed decreased beating rate, beat-to-beat and Ca^{2+} flux manifested severe abnormalities in neonatal rat derived primary cardiomyocytes. In the case of immunotherapies, tyrosine kinase inhibitors (TKI) were investigated the most (Xian et al., 2022). In a study using transparent MEAs, both structural (with HCA assay) and functional (electrophysiology and contractility analysed with MEA) effect of a panel of TKIs was investigated. In another study, the cardiotoxicity of a novel third generation TKI, AC0010 was investigated using a bifunctional MEA (Jiang et al., 2023).

In the Table 5 several publications on the use of MEAs for cardiotoxicity assessment have been listed and the drugs, the MEA set up, the used cell type and studies parameters during recording are mentioned in the Table 5.

Conclusion and future outlook

The revolution in cancer treatment has significantly increased the number of cancer survivors and especially in the paediatric and geriatric population. Unfortunately, these cancer patients are now faced with the acute and chronic adverse effects of cancer treatments. One of the most notable cancer treatments' adverse effects is cardiotoxicity. Cancer treatment induced cardiotoxicity (CIC) can be studied in more depth *in vitro* which helps with understanding its underlying mechanism and aids in developing cardioprotective drugs and heart-friendly cancer therapies.

One of the *in vitro* methods for studying CIC non-destructively and in real time is bioelectric analysis with microelectrode arrays (MEAs). MEAs have evolved significantly over the past 50 years since the first successful measurements of chick myocytes by Thomas et al. (1972). Traditionally, high-throughput screening of cardiac tissue requires different platforms to evaluate both electrical as well as mechanical properties, which often limits the correlation between interrelated physiological events. Modern multi-modal microelectrode array (MEA) platforms address this issue by integrating multiple (distinct) sensing modalities onto a single substrate. This enables the simultaneous evaluation of the structural, electrophysiological and contractile characteristics of cardiomyocytes (Miccoli et al., 2018). Electrophysiological features are captured via extracellular field potential recordings using planar arrays of up to 500 microelectrodes in combination with amplifier electronics. A higher spatial signal resolution with several 1,000 electrodes can be achieved with high-density CMOS

TABLE 5 List of recent and pioneering works using MEAs for cardiotoxicity assessment by pharmaceuticals and cancer treatments.

#	Title	Drugs	Parameters	MEA and MEA setup	Cell type	References
1	Micro-Electrode Arrays in Cardiac Safety Pharmacology	E4031, amiodarone, quinidine and sotalol	fAPD prolongation, EADs	MEA1060 System, two types of MEA, type 1 (30 microns in diameter and 200 microns apart – titanium nitride), type 2 (100 microns in diameter, 700 microns apart, gold). 60 electrodes. MEA and MEA setup from Multichannel systems.	embryonic chicken ventricular cells	Meyer et al. (2004)
2	Screening drug-induced arrhythmia	Torsadogenic hERG blockers, such as sotalol and quinidine, a comprehensive panel of drugs	fAPD, Beat rate, EAD, Ectopic beats	AL-MED P5004A MEA probes (Alpha Med Scientific, Osaka, Japan)	human embryonic stem cell-derived cardiomyocytes hESC-CMs formed as embryoid bodies used at 65 and 95 days postinduction	Navarrete et al. (2013)
3	Multi-chamber cardioids unravel human heart development and cardiac defects	—	FP recording using CMOS-MEA	CMOS_MEA from 3Brain	Multi-chambered cardioids	Schmidt et al. (2023)
4	Comprehensive *in vitro* cardiac safety assessment using human stem cell technology: Overview of CSAHi HEART initiative	E-4031, Flecainide, Terfenadine, Moxifloxacin, Chromanol293B	FP corrected were measured by the Maestro, MED64 and MEA60/120. IBD10 was measured with xCELL.	MED64, MEA60/2100, Maestro, xCELL	Human iPSC derived cardiomyocytes	Xian et al. (2022)
5	Cardiomyocyte functional screening: Interrogating comparative electrophysiology of high-throughput model cell systems	—	FP recording	MEA with 60 electrodes with a layer of conductive polymer on top of the electrodes (from multichannel systems).	Primary neonatal rat ventricular myocytes, immortalized mouse atrial HL-1 cell line and human induced pluripotent stem-cell derived cardiomyocytes.	Wells et al. (2019)
6	Electromechanical integrated recording of single cardiomyocyte *in situ* by multimodal microelectrode biosensing system	—	FP recording and contraction	MEAs on glass substrate with 32 electrodes and 10 microns in diameter were fabricated in house. The glass with electrodes is attached to a PCB board with pads.	Neonatal rat cardiomyocytes culture	Zhang et al. (2022)
7	Intracellular recording of cardiomyocyte action potentials by nano branched microelectrode array	—	Intracellular like recording	Nanobranches are fabricated a standard glass slide as mentioned in the reference.	Primary rat cardiomyocyte culture	Hu et al. (2020)

(*Continued*)

TABLE 5 Continued

#	Title	Drugs	Parameters	MEA and MEA setup	Cell type	References
8	A universal, multimodal cell-based biosensing platform for optimal intracellular action potential recording	—	Intracellular like recording	MEA with an electroporation setup with differential control and adjustment of the electroporation parameter (amplitude, pulse width, pulse number of the electroporating pulses).	Primary neonatal rat cardiomyocytes	Xu et al. (2022)
9	Intracellular recording of action potentials by nanopillar electroporation	—	Intracellular like recording and extracellular electrodes	Vertical nanopillar electrodes and repeated nanoscale electroporation and resealing processes and MEAs.	HL-1 cells	Xie et al. (2012)
10	Simultaneous electrical recording of cardiac electrophysiology and contraction on chip	Nifedipine	FP and contractility	MEA and IDES microelectrodes were patterned on glass using photolithography.	Human iPSC-CMs	Qian et al. (2017)
11	Cardiotoxicity assessment through a polymer-based cantilever platform: An integrated electro-mechanical screening approach	Nifedipine, Bay K-8644, Astemizole, Quinidine, Blebbistatin, Isoproterenol	FP duration, contractility.	Eight cantilevers with patterned strain sensors and MEA	Primary neonatal rat cardiomyocytes	Kanade et al. (2024)
12	Modeling doxorubicin-induced-cardiotoxicity through breast cancer patient specific iPSC-derived heart organoid	Doxorubicin	FP and beating rate period	n/a	Breast cancer patient derived heart organoids with doxorubicin induced cardiotoxicity and without	Jang et al. (2024)
13	High-temporal-range drug-induced cardiac side effect evaluation using simultaneous HL-1 based impedance and long-term electrophysiology recording systems	Vinblastine and nifedipine	FP duration and contractility	The setup has 8 wells and each well has a pair of IDEs, two planar microelectrodes in the center and reference electrodes aside.	HL-1 cell line	Gu et al. (2019)
14	Evaluation of doxorubicin toxicity on cardiomyocytes using a dual functional extracellular biochip	doxorubicin	Extracellular field potentials and cell–substrate impedance	ExCell for detection of both extracellular potential and impedance in an assay	The hearts of 1- to 3-day-old neonatal Sprague Dawley rats were used	Xiao et al. (2010)
15	Long-term *in vitro* recording of cardiac action potentials on microelectrode arrays	doxorubicin	Intracellular-like action potentials	Optoporation and transparent (Titanium nitride) 6 well MEAs from multichannel systems	hiPSC-derived cardiomyocytes purchased from Dynamics International and Ncardia	Xu et al. (2022)
16	Electro-metabolic coupling in multi-chambered vascularized human cardiac organoids	Mitoxantrone	FP recording, real time contraction and oscillation of oxygen	Custom designed transparent 9 well MEA	Multichambered vascularized cardioids	Qin et al. (2024)

(*Continued*)

TABLE 5 Continued

#	Title	Drugs	Parameters	MEA and MEA setup	Cell type	References
17	High-throughput assessment of drug cardiac safety using high-speed impedance detection technology-based heart-on-a-chip.	Doxorubicin, Verapamil	Beating rate and amplitude derived from mechanical contraction signals	The MEAs with 16 IDEs with 5 mm in diameter, 9 mm in center to center spacing and with 110-micron distance between IDEs branches were used. A polystyrene chamber was fixed on the defined IDEs.	Primary neonatal rat cardiomyocytes	Zhang et al. (2016)
18	Evaluating the efficacy and cardiotoxicity of EGFR-TKI AC0010 with a novel multifunctional biosensor	AC0010 is a third generation Tyrosine kinase inhibitor	Cell viability, contraction, FP recording	The MEAs contained IDEs for contraction and viability monitoring with two field potential recording electrodes	Primary neonatal rat cardiomyocytes	Jiang et al. (2023)
19	Efficacy and cardiotoxicity integrated assessment of anticancer drugs by a dual functional cell -based biosensor	Taxol and vinblastine	FP and contractility	The MEAs contained IDEs for contraction and viability monitoring with two field potential recording electrodes	Primary neonatal rat cardiomyocytes and Hela cells.	Wei et al. (2019)
20	Bridging functional and structural cardiotoxicity assays using human embryonic stem cell derived cardiomyocytes for a more comprehensive risk assessment	Amiodarone, Aspirin, Astemizole, Axitinib, AZT, Bepridil, Doxorubicin, E-4031, Mexiletine, Rosiglitazone, Sunitinib, Sibutramine and Verapamil.	Functional and structural cardiotoxicity parameters	Transparent MEA from Axion biosystems and Cytvia cell health assay for HCA kit	hiPSC-derived cardiomyocytes	Clements et al. (2015)
21	Evaluation of doxorubicin toxicity on cardiomyocytes using a dual functional extracellular biochip	Doxorubicin	FP, physical state of CMs and contractility	ExCell	Primary neonatal rat cardiomyocytes	Xiao et al. (2010)
22	Kinase inhibitor-induced cardiotoxicity assessed *in vitro* with human pluripotent stem cell derived cardiomyocytes	Axitinib, sunitinib malate, temsirolimus, Alcocidib, barasertib, selumetinib, XL765, dasatinib, erlotinib, everolimus, imatinib, lapatinib, lestaurtinib, tozasertib.	*In vitro* arrhythmic events (iAE), field potential duration (FPD) prolongation and spike amplitude suppression (SAS).	Maestro MEA system with AxIS software (Axion Biosystems) was used.	Human iPSC derived cardiomyocytes	Xian et al. (2022)

microelectrodes where amplifiers and filter circuitry are embedded within the same chip together with the microelectrodes. If not, only extracellular field potentials shall be recorded but membrane potentials and intracellular voltages, then 3-dimensional nanopillar MEAs often in combination with electroporation or with mechanical force, are used. Potential recording with MEAs delivers high temporal resolution data on cardiac rhythm, conduction velocity and ion channel kinetics across large cellular networks (Xu et al., 2022). At the same time, the contractility of cardiomyocytes can be quantified through real-time impedance measurements using the same (or comparable) microelectrode arrayshese impedance measurements carry the subtle variations in electrical resistance at the electrode-electrolyte interface that are caused by the cells' rhythmic morphological changes and detachment during physical contraction (Miccoli et al., 2018; Bounik et al., 2022). As these advanced chips can be engineered with optically transparent conductive electrode materials or coupled directly with optical imaging frameworks, this dual-function electrical monitoring can be seamlessly integrated with high-resolution structural assays (Miccoli et al., 2018; Lee et al., 2026). This comprehensive on-chip integration enables researchers to correlate structural phenotype directly with functional excitation-contraction coupling in a unified, non-invasive environment. Building on this multi-modal capability, it is also possible to

non-invasively monitor the effects of the cancer treatments on both cancer cells and cardiomyocytes on the same chip. Here, impedance measurements track both cancer cell viability and cardiomyocytes contractility in tandem. This integrated *in vitro* approach accelerates the field of precision cardio-oncology, translating it from clinical settings to *in vitro* personalized, cancer therapeutic screening. Furthermore, while traditional MEAs are limited to recording weaker extracellular signals, the introduction of transient micro-electroporation protocols has a strong impact in their applications. By applying localized, low voltage electrical pulses, researchers can temporarily permeabilize the cell membrane to enable intracellular-like measurement of action potentials on planar MEAs without compromising cell viability (Fendyur and Spira, 2012). Combining this intracellular resolution with extracellular field potentials and structural assays yields a truly comprehensive, non-invasive, label free cardiotoxicity assessment assay. Ultimately, these advanced MEA chips offer a powerful strategy for investigating the multi-faceted mechanisms of cancer treatment induced cardiotoxicity on a single, unified platform.

Moving forward, the next-generation of MEA-based cardiotoxicity screening will be driven by the convergence of scaled hardware automation and advanced computational workflows. The necessity to enhance assay throughput remains paramount, necessitating a transition from conventional low-well formats to high-density configurations, such as 96- and 384-well plates, which are capable of expeditious, parallel compound screening (Belbachir et al., 2022). In order to match this expanded hardware capacity, it is essential to integrate automated cell culture platforms and robotic drug-dispensing workflows. This will minimise human error, reduce batch-to-batch variability in human induced pluripotent stem cell-derived cardiomyocyte (hiPSC-CM) cultures, and enable continuous, long-term kinetic monitoring (Isobe et al., 2019; Zhao et al., 2022). It is imperative to acknowledge the substantial data streams generated by these high-throughput, multiparametric recordings, which will necessitate a transition from manual feature extraction towards the utilisation of artificial intelligence (AI) and machine learning-assisted data analysis (Kalkunte et al., 2024). The employment of deep learning architectures and automated signal-processing algorithms has been demonstrated to facilitate the rapid classification of complex waveform anomalies, the prediction of proarrhythmic risks, and the detection of subtle, multi-channel field potential dynamics that evade conventional analysis. This has led to the establishment of MEAs as robust, fully automated discovery platforms in safety pharmacology (Kowalczewski et al., 2022).

Author contributions

NK: Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Software, Validation, Visualization, Writing – original draft, Writing – review and editing. SK: Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Software, Validation, Visualization, Writing – original draft, Writing – review and editing. HW: Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Software, Validation, Visualization, Supervision, Writing – original draft, Writing – review and editing. SP-L: Conceptualization, Data curation, Formal Analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Software, Supervision, Validation, Visualization, Writing – original draft, Writing – review and editing.

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Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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